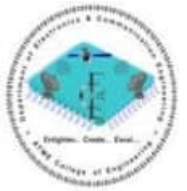




ATME
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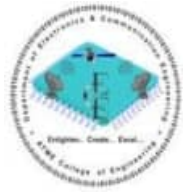
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Module 4

Point Sources and Arrays & Electric Dipoles

Module 4: Chapter I

Point Sources and Arrays:

- Introduction
- Point Sources
- Power Patterns
- Power Theorem
- Radiation Intensity
- Field Patterns
- Phase Patterns
- Arrays of Two Isotropic Point Sources
- Pattern Multiplication
- Linear Arrays of n Isotropic Point Sources of equal Amplitude and Spacing

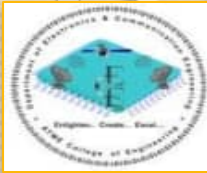
Suggested Reading

Text 3: “Antennas and Wave Propagation”

John D. Krauss, Ronald J Marhefka and Ahmad S Khan, 4th
Special Indian Edition , McGraw- Hill Education Pvt. Ltd., 2010.

Text 3:

Sections: 5.1 – 5.11, 5.13

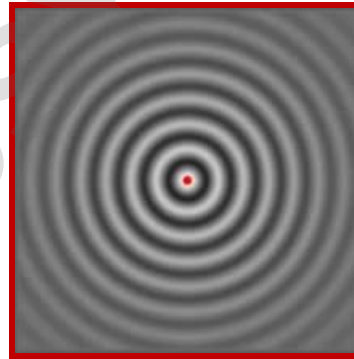


Introduction

Point Sources:

Definition: Point sources in radiation refer to localized sources *emitting radiation from a single point*.

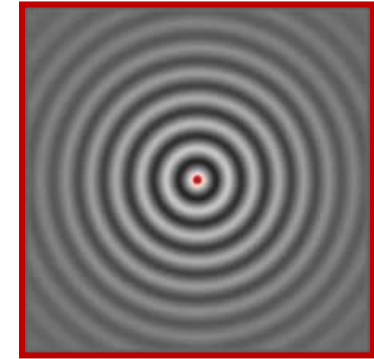
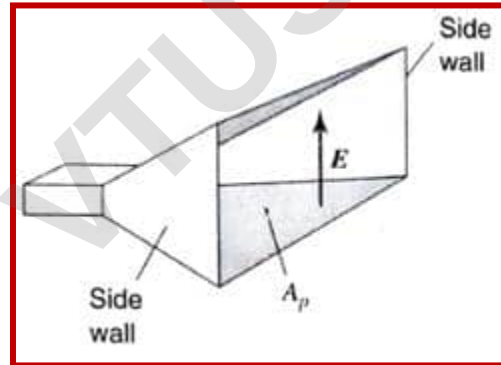
- These sources are considered to be *infinitesimally small* and radiate energy uniformly *in all directions*.
- The concept of a point source is often used in antenna theory and electromagnetic analysis to simplify the study and design of more complex antenna systems.



5-1 Introduction

In chap. 2 an antenna was treated as an aperture. In this chapter an antenna is first considered as a point source and later the concept is extended to the formation of arrays of point sources. This approach is of great value since **the pattern of any antenna can be regarded as produced by an array of point sources**. Further the initial discussion relating to arrays is confined to isotropic point sources, which may represent different kinds of antennas. Later the discussion is extended to encompass more general case of non-uniform distribution.

Radio wave sources which are smaller than one radio wavelength are also generally treated as point source.



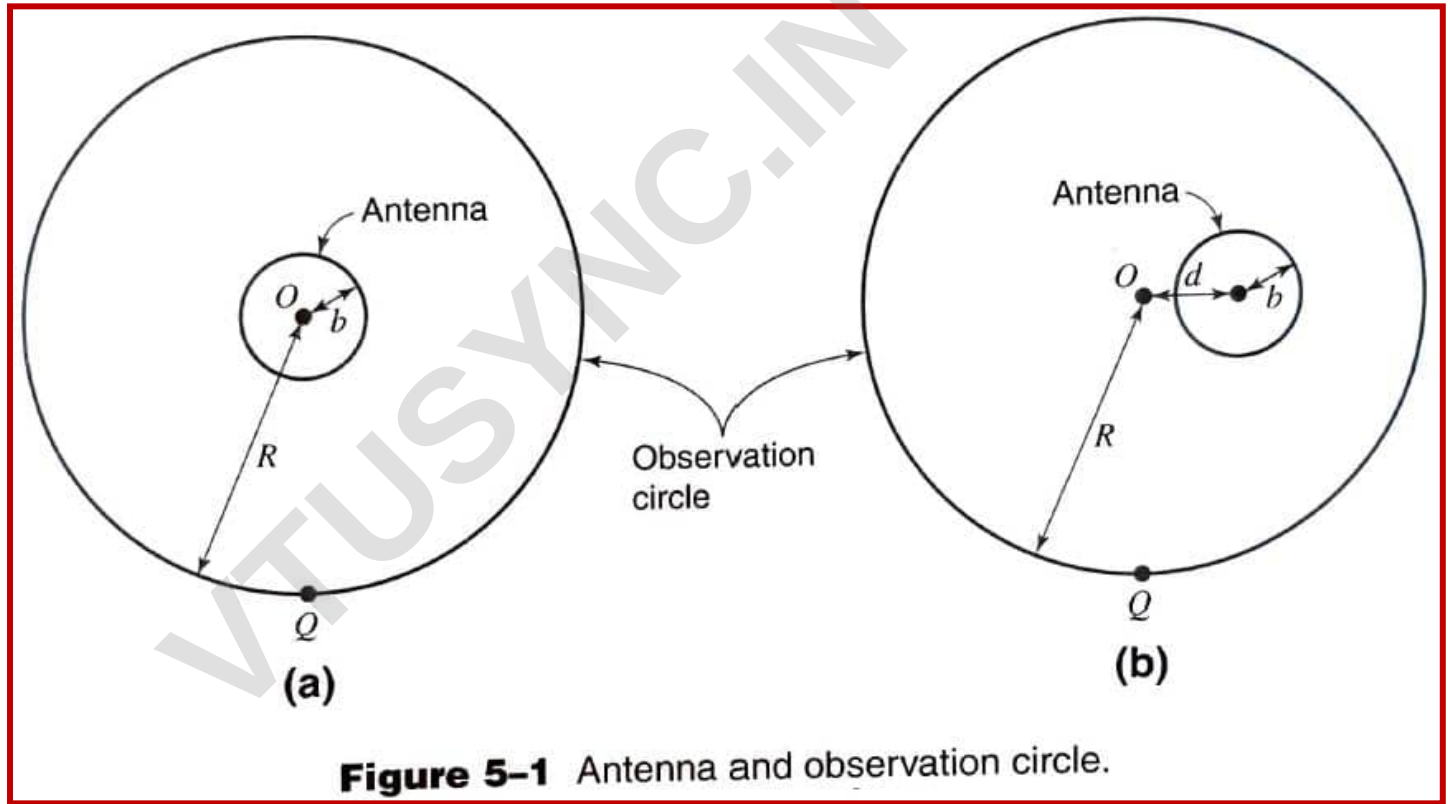
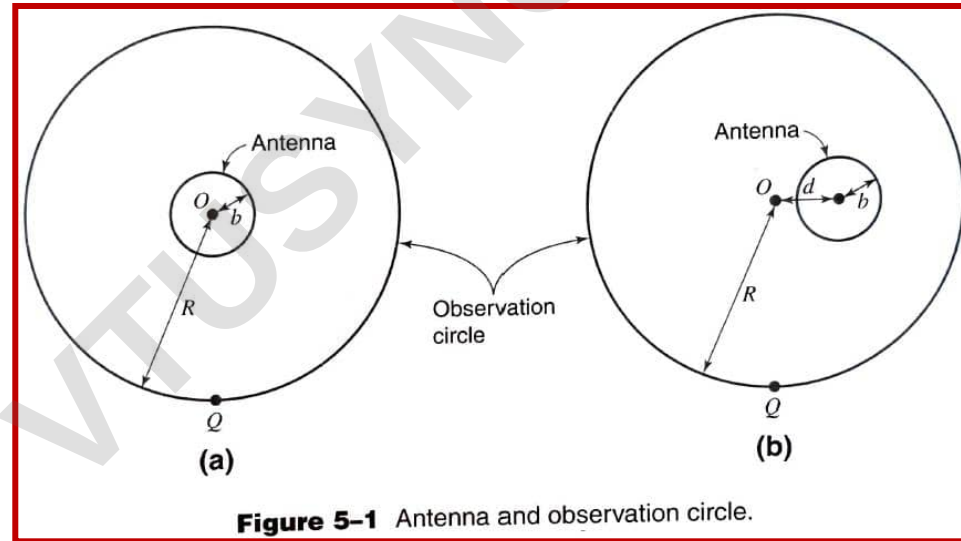


Figure 5-1 Antenna and observation circle.

At a sufficient distance in the far field of an antenna, the radiated fields of the antenna are transverse and the power flow or Poynting vector (Wm^{-2}) is radial as at the point O at a distance R on the observation circle in Fig. 5–1. It is convenient in many analyses to assume that the fields of the antenna are everywhere of this type. In fact, we may assume, by extrapolating inward along the radii of the circle, that the waves originate at a fictitious volumeless emitter, or *point source*, at the center O of the observation circle.

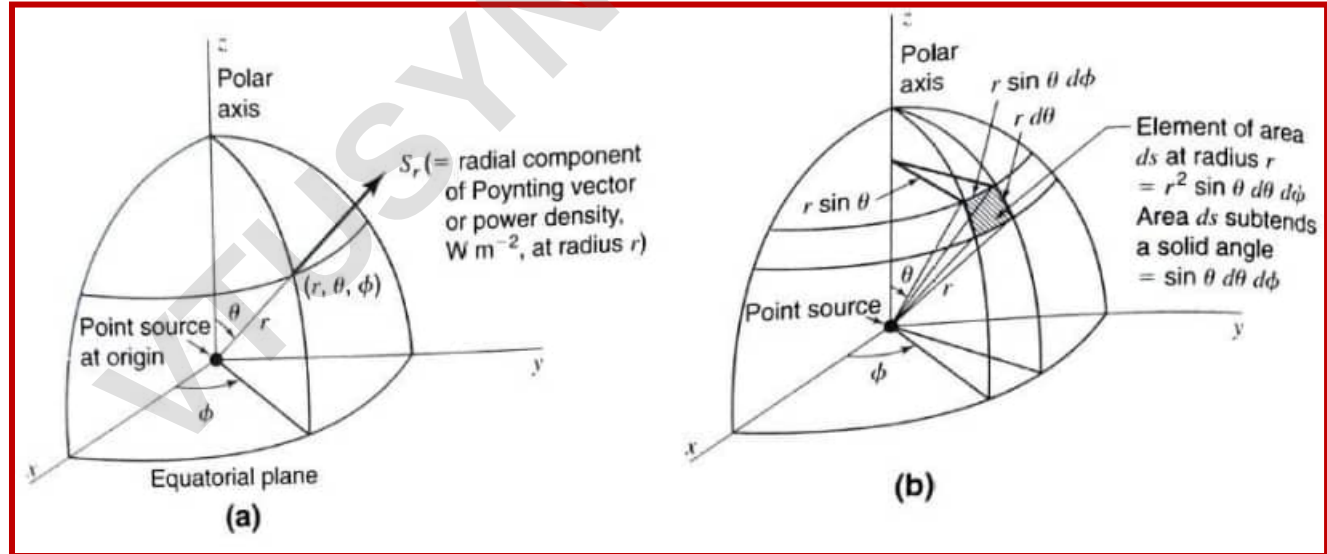
The actual field variation near the antenna, or "near field," is ignored, and we describe the source of the waves only in terms of the "far field" it produces. Provided that our observations are made at a sufficient distance, any antenna, regardless of its size or complexity, can be represented in this way by a single point source.

In Fig. 5-1a, the center O of the antenna coincides with the center of the observation circle. If the center of the antenna is displaced from O , even to the extent that O lies outside the antenna as in Fig. 5-1b, the distance d between the two centers has a negligible effect on the field patterns at the observation circle, provided $R \gg d$, $R \gg b$, and $R \gg \lambda$. However, the phase patterns will generally differ, depending on d . If $d = 0$, the phase shift around the observation circle is usually a minimum. As d is increased, the observed phase shift becomes larger.



Power Patterns

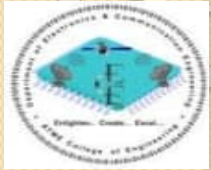
Let a transmitting antenna in free space be represented by a point-source radiator located at the origin of the coordinates in Fig. 5–2 (see also Fig. 2–5). The radiated energy streams from the source in radial lines. The coordinates in Fig. 5–2 (see also Fig. 2–5). The radiated energy streams from the source in radial lines. The time rate of energy flow per unit area is the *Poynting vector*, or *power density* (watts per square meter). For a point source (or in the far field of any antenna), the Poynting vector $\mathbf{S}$ has only a radial component S_r with no components in either the θ or ϕ directions ($S_\theta = S_\phi = 0$). Thus, the magnitude of the Poynting vector, or power density, is equal to the radial component ($|\mathbf{S}| = S_r$).





Types of point sources:

- Isotropic Source
- Anisotropic Source



A source that radiates energy uniformly in all directions is an *isotropic source*. For such a source the radial component S_r of the Poynting vector is independent of θ and ϕ . A graph of S_r at a constant radius as a function of angle is a Poynting vector, or power-density, pattern, but is usually called a *power pattern*. The three-dimensional power pattern for an isotropic source is a sphere. In two dimensions the pattern is a circle (a cross section through the sphere), as suggested in Fig. 5–3.

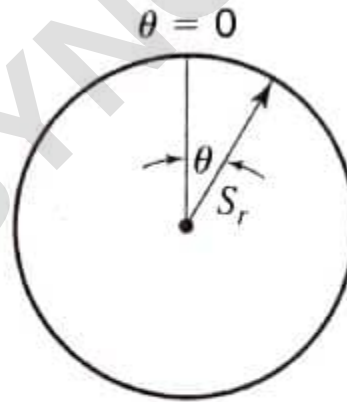
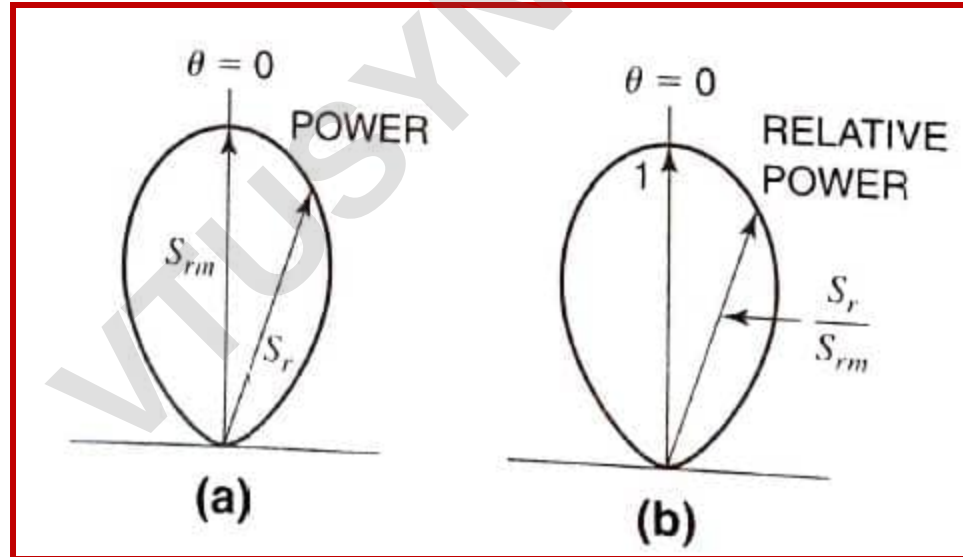


Figure 5–3 Polar power pattern of isotropic source.

If S_r is expressed in watts per square meter, the graph is an *absolute power pattern*. On the other hand, if S_r is expressed in terms of its value in some reference direction, the graph is a *relative power pattern*. It is customary to take the reference direction such that S_r is a maximum. Thus, the pattern radius for relative power is S_r/S_{rm} where S_{rm} is the maximum value of S_r . The maximum value of the relative power pattern is unity, as shown in Fig. 5-4b. A pattern with a maximum of unity is also called a *normalized pattern*.



A Power Theorem' and its Application to an Isotropic Source:

CBCS SCHEME

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15EC71

Seventh Semester B.E. Degree Examination, Dec.2019/Jan.2020

Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

ice.

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-4

- 7 a. Prove that directivity for a source with unidirectional pattern of $U_m \cos^n \theta$, where 'n' can be any number, can be expressed as $D = 2(n + 1)$. (06 Marks)
- b. Obtain field expression of two isotropic point sources of same amplitude and phase. (10 Marks)

OR

- 8 a. State and explain power theorem. (06 Marks)
- b. Derive an expression for radiation resistance of short electric dipole. (10 Marks)

If the Poynting vector is known at all points on a sphere of radius r from a point source in a lossless medium, the total power radiated by the source is the integral over the surface of the sphere of the radial component S_r of the average Poynting vector. Thus,

$$P = \oiint S \cdot ds = \oiint S_r ds \quad (1)$$

where

P = power radiated, W

S_r = radial component of average Poynting vector, W m^{-2}

ds = infinitesimal element of area of sphere (see Fig. 3-2b)
 $= r^2 \sin \theta d\theta d\phi$, m^2

For an isotropic source, S_r is independent of θ and ϕ so

$$P = S_r \oiint ds = S_r \times 4\pi r^2 \quad (\text{W}) \quad (2)$$

and

$$S_r = \frac{P}{4\pi r^2} \quad (\text{W m}^{-2}) \quad (3)$$

Equation (3) indicates that the magnitude of the Poynting vector varies inversely as the square of the distance from a point-source radiator. This is a statement of the well-known law for the variation of power per unit area as a function of the distance.

Radiation Intensity

As discussed in Sec. 2–5, the radiation intensity U is expressed in watts per unit solid angle ($W \text{ sr}^{-1}$). The radiation intensity is independent of radius. It is power per steradian. From (5–4–3) we have

$$r^2 S_r = P/4\pi = U \quad (W/\text{sr}) \quad (1)$$

Thus, the power theorem may be restated as follows:

The total power radiated is given by the integral of the radiation intensity over a solid angle of 4π steradians.

Applying (1) to an isotropic source gives

$$P = 4\pi U_0 \quad (W)$$

where U_0 = radiation intensity of isotropic source, $W \text{ sr}^{-1}$.

EXAMPLE 5-6.1 Source with Unidirectional Cosine Power Pattern

A source has a cosine radiation-intensity pattern, that is,

$$U = U_m \cos \theta$$

where U_m = maximum radiation intensity

The radiation intensity U has a value only in the upper hemisphere ($0 \leq \theta \leq \pi/2$ and $0 \leq \phi \leq 2\pi$) and is zero in the lower hemisphere. The radiation intensity is a maximum at $\theta = 0$. The pattern is shown in Fig. 5-5. The space pattern is a figure of revolution of this circle around the polar axis. Find the directivity.

$$P = \int_0^{2\pi} \int_0^{\pi/2} U_m \cos \theta \sin \theta \, d\theta \, d\phi$$

To find the total power radiated by the cosine source, we apply (5-4-1) and integrate only over the upper hemisphere. Thus

$$P = \int_0^{2\pi} \int_0^{\pi/2} U_m \cos \theta \sin \theta d\theta d\phi = \pi U_m \quad (2)$$

If the power radiated by the unidirectional cosine source is the same as for an isotropic source, then (2) and (1) in Sec. 5-5 may be set equal, yielding

$$\pi U_m = 4\pi U_0$$

or

$$\text{Directivity} = \frac{U_m}{U_0} = 4 = D \quad \text{Ans} \quad (3)$$

Thus, the maximum radiation intensity U_m of the unidirectional cosine source (in the direction $\theta=0$) is 4 times the radiation intensity U_0 from an isotropic source radiating the same total power. The power patterns for the two sources are compared in Fig. 5-6 for the same total power radiated by each.

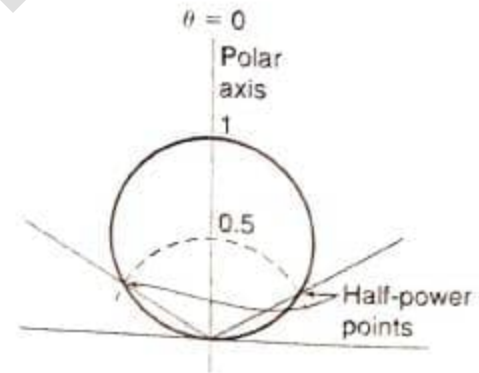


Figure 5-5 Unidirectional cosine power pattern

EXAMPLE 5-6.2 Source with Bidirectional Cosine Power Pattern

A source has a cosine power pattern that is bidirectional. Find the directivity. With radiation in two hemispheres instead of one; the maximum radiation intensity is half its value in Example 5-6.1. Thus, from (3),

$$D = 4/2 = 2 \quad \text{Ans.}$$

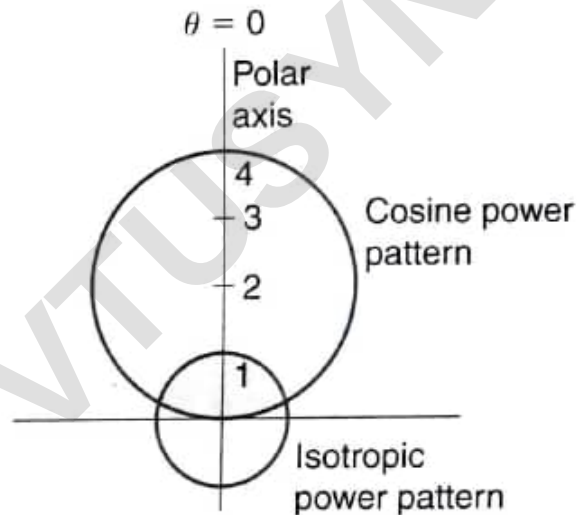


Figure 5-6
Power patterns of
cosine and
isotropic sources.

EXAMPLE 5-6.3 Source with Sine (Doughnut) Power Pattern

A source has a radiation intensity pattern given by

$$U = U_m \sin \theta$$

The pattern is shown in Fig. 5-6. The space pattern is a figure-of-revolution of this pattern around the polar axis and has the form of a doughnut. Find D .

$$P = U_m \int_0^{2\pi} \int_0^{\pi} \sin^2 \theta \, d\theta \, d\phi$$

■ Solution

Applying (5-4-3) the total power radiated is

$$P = U_m \int_0^{2\pi} \int_0^\pi \sin^2 \theta \, d\theta \, d\phi = \pi^2 U_m$$

If the power radiated by this source is the same as for an isotropic source taken as reference, we have

$$\pi^2 U_m = 4\pi U_0$$

and

$$\text{Directivity} = \frac{U_m}{U_0} = \frac{4}{\pi} = 1.27 = D \quad \text{Ans.}$$

EXAMPLE 5-6.4 Source with Sine-Squared (Doughnut) Power Pattern

A source has a sine-squared radiation-intensity power pattern. The radiation-intensity pattern is given by

$$U = U_m \sin^2 \theta$$

$$P = U_m \int_0^{2\pi} \int_0^\pi \sin^3 \theta \, d\theta \, d\phi$$

$$\begin{aligned}\int \sin^3(x) dx &= \int \sin(x)(1 - \cos^2(x)) dx \\ &= \int \sin(x) dx - \int \sin(x)\cos^2(x) dx\end{aligned}$$

For the first integral:

$$\int \sin(x) dx = -\cos(x) + C$$

For the second integral, using [substitution](#):

$$\text{Let } u = \cos(x) \Rightarrow du = -\sin(x) dx$$

Then

$$-\int \sin(x)\cos^2(x) dx = \int u^2 du$$

$$= \frac{u^3}{3} + C$$

$$= \frac{1}{3}\cos^3(x) + C$$

Putting it all together, we get our final result:

$$\begin{aligned}\int \sin^3(x) dx &= \int \sin(x) dx - \int \sin(x)\cos^2(x) dx \\ &= -\cos(x) + \frac{1}{3}\cos^3(x) + C\end{aligned}$$

$$\int \sin^3(x) dx = \frac{1}{3}\cos^3(x) - \cos(x) + C$$

The power pattern is shown in Fig. 5-7a. This type of pattern is of considerable interest because it is the pattern produced by a short dipole coincident with the polar ($\theta = 0$) axis in Fig. 5-7a. Applying (5-4-3), the total power radiated is

$$P = U_m \int_0^{2\pi} \int_0^\pi \sin^3 \theta \, d\theta \, d\phi = \frac{8}{3} \pi U_m$$

If P is the same as for the isotropic source,

$$\frac{8}{3} \pi U_m = 4\pi U_0$$

and

$$\text{Directivity} = \frac{U_m}{U_0} = \frac{3}{2} = 1.5 = D \quad \text{Ans.}$$

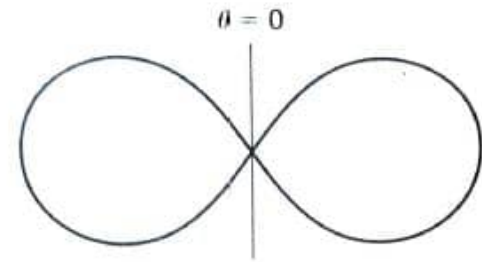


Fig. 5-7a.

EXAMPLE 5-6.5 Source with Unidirectional Cosine-Squared Power Pattern

A source with a unidirectional cosine-squared radiation-intensity pattern is given by

$$U = U_m \cos^2 \theta$$

The radiation intensity has a value only in the upper hemisphere as in Fig. 5-7b. The three-dimensional or space pattern is a figure-of-revolution of this pattern around the polar ($\theta = 0$) axis. Find the directivity.

$$P = U_m \int_0^{2\pi} \int_0^{\pi/2} \cos^2 \theta \sin \theta d\theta d\phi$$

$$\int \cos^2 x \sin x dx$$

First remember that $\cos^2 x = (\cos x)^2$, so we have:

$$\int (\cos x)^2 \sin x dx.$$

Use substitution:

Let $u = \cos x$, so that $du = -\sin x dx$.

$$\int \cos^2 x \sin x dx = - \int u^2 du$$

The total power radiated is

The total power radiated is

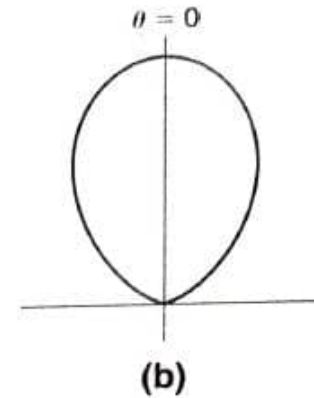
$$P = U_m \int_0^{2\pi} \int_0^{\pi/2} \cos^2 \theta \sin \theta \, d\theta \, d\phi = \frac{2}{3} \pi U_m$$

If P is the same as radiated by an isotropic source,

$$\frac{2}{3}\pi U_m = 4\pi U_0$$

and

$$\text{Directivity} = \frac{U_m}{U_0} = 6 = D \quad \text{Ans.}$$



Thus, the maximum power per unit solid angle (at $\theta = 0$) from the source with the cosine-squared power pattern is six times the power per unit solid angle from an isotropic source radiating the same power.

Table 5-1

Directivities of the Point Source Patterns in Examples 5-6.1 to 5-6.5.

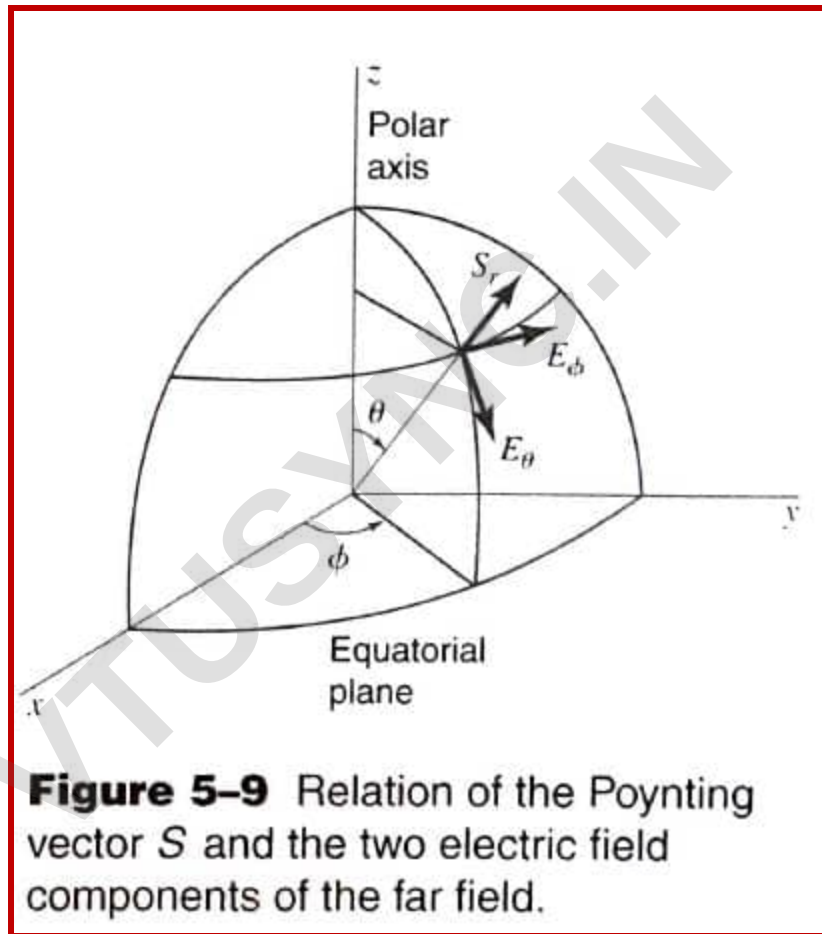
Pattern	Directivity
Unidirectional cosine	4
Bidirectional cosine	2
Sine doughnut	1.27
Sine-squared doughnut	1.5
Unidirectional cosine squared	6

Field Patterns

The discussion in the preceding sections is based on considerations of power. This has afforded a simplicity of analysis, since the power flow from a point source has only a radial component which can be considered as a scalar quantity. To describe the field of a point source more completely, we need to consider the electric field $\mathbf{E}$ and/or the magnetic field $\mathbf{H}$ (both vectors). For point sources we deal entirely with *far fields* so $\mathbf{E}$ and $\mathbf{H}$ are both entirely transverse to the wave direction, are perpendicular to each other, are in-phase, and are related in magnitude by the intrinsic impedance of the medium ($\mathbf{E}/\mathbf{H} = Z = 377 \Omega$ for free space). For our purposes it suffices to consider only one field vector, and we arbitrarily choose the electric field $\mathbf{E}$.

Since the Poynting vector around a point source is everywhere radial, it follows that the electric field is entirely transverse, having only E_θ and E_ϕ components. The relation of the radial component S_r of the Poynting vector and the electric field components is illustrated by the spherical coordinate diagram of Fig. 5–9. The conditions characterizing the far field are then:

1. Poynting vector radial (S_r component only)
2. Electric field transverse (E_θ and E_ϕ components only)



The relation between the average Poynting vector and the electric field at a point of the far field is

$$S_r = \frac{1}{2} \frac{E^2}{Z}$$

where Z_0 = intrinsic impedance of medium and

$$E = \sqrt{E_\theta^2 + E_\phi^2}$$

where

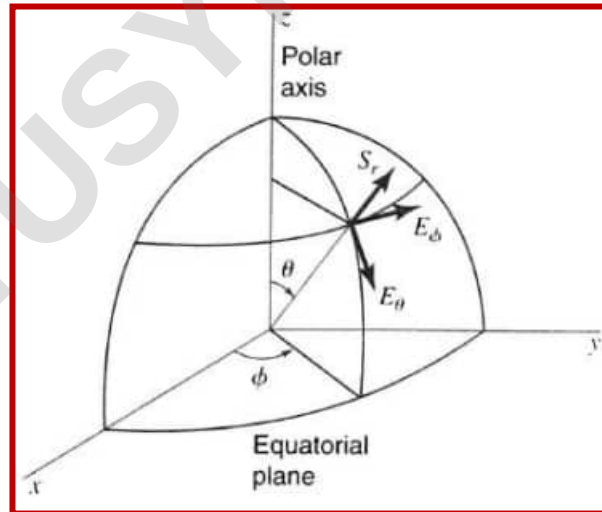
E = amplitude of total electric field intensity

E_θ = amplitude of θ component

E_ϕ = amplitude of ϕ component

The field may be elliptically, linearly or circularly polarized.

A pattern showing the variation of the electric field intensity at a constant radius r as a function of angle (θ, ϕ) is called a *field pattern*. In presenting information concerning the far field of an antenna, it is customary to give the field patterns for the two components, E_θ and E_ϕ , of the electric field since the total electric field $\mathbf{E}$ can be obtained from the components by (2), but the components cannot be obtained from a knowledge of only $\mathbf{E}$.



When the field intensity is expressed in volts per meter, it is an *absolute field pattern*.¹ On the other hand, if the field intensity is expressed in units relative to its value in some reference direction, it is a *relative field pattern*. The reference direction is usually taken in the direction of maximum field intensity. The relative pattern of the E_θ component is then given by

$$\frac{E_\theta}{E_{\theta m}}$$

and the relative pattern of the E_ϕ component is given by

$$\frac{E_\phi}{E_{\phi m}}$$

where

$E_{\theta m}$ = maximum value of E_θ

$E_{\phi m}$ = maximum value of E_ϕ

The magnitudes of both the electric field components, E_θ and E_ϕ , of the far field vary inversely as the distance from the source. However, they may be different functions, F_1 and F_2 , of the angular coordinates, θ and ϕ . Thus, in general,

$$E_\theta = \frac{1}{r} F_1(\theta, \phi)$$

$$E_\phi = \frac{1}{r} F_2(\theta, \phi)$$

Since $S_{rm} = E_m^2/2Z$, where E_m is the maximum value of E , it follows on dividing this into (1) that the relative total power pattern is equal to the square of the relative total field pattern. Thus,

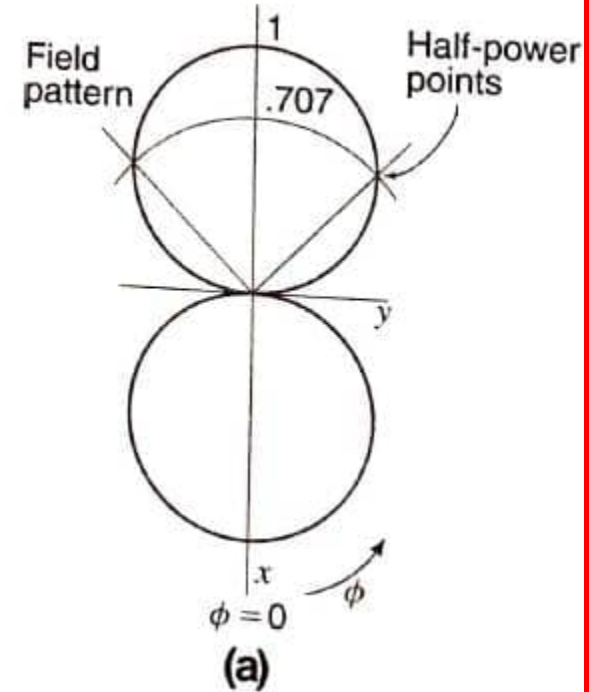
$$P_n = \frac{S_r}{S_{rm}} = \frac{U}{U_m} = \left(\frac{E}{E_m} \right)^2$$

EXAMPLE: Source with Cosine Field Pattern

An antenna's far field has only an E_ϕ component in the equatorial plane, the E_θ component being zero in this plane. The relative equatorial-plane pattern of the E_ϕ component (that is, E_ϕ as a function of ϕ for $\theta = 90^\circ$) is given by

$$\frac{E_\phi}{E_{\phi m}} = \cos \phi$$

Find D .



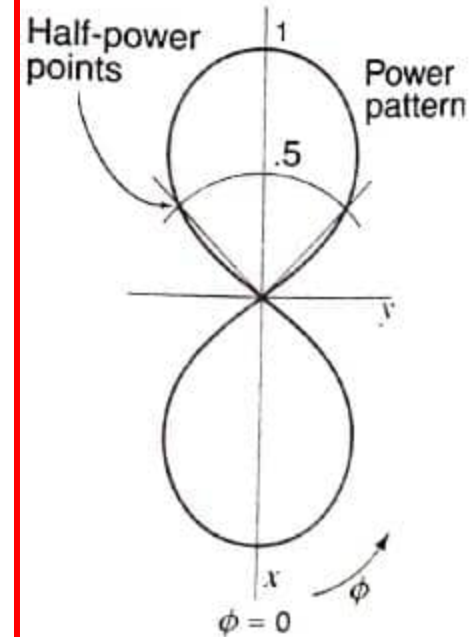
Solution:

The relative (normalized) power pattern in the equatorial plane is equal to the square of the relative field pattern. Thus

$$P_n = \frac{S_r}{S_{rm}} = \frac{U}{U_m} = \left(\frac{E_\phi}{E_{\phi m}} \right)^2$$

and substituting

$$P_n = \cos^2 \phi$$

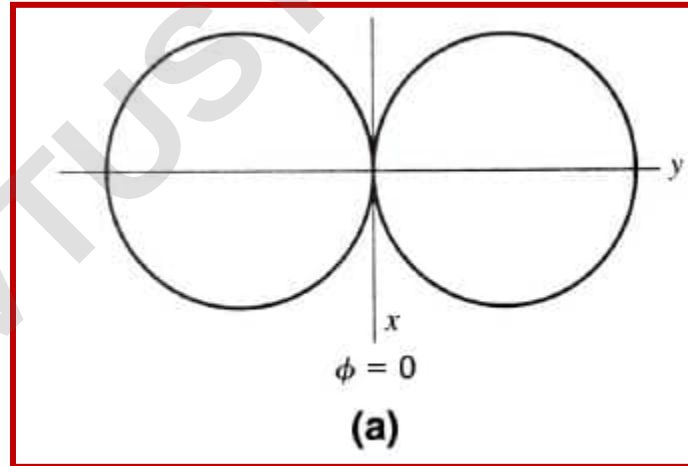


EXAMPLE 5-7.2 Source with Sine Field Pattern

An antenna has a far field that has only an E_θ component in the equatorial plane, the E_ϕ component being zero in this plane. Assume that the relative equatorial-plane pattern of the E_θ component (that is, E_θ as a function of ϕ for $\theta = 90^\circ$) for this antenna is given by

$$\frac{E_\theta}{E_{\theta m}} = \sin \phi$$

This pattern is illustrated by Fig. 5-11a and could be produced by a small loop antenna, the axis of the loop coincident with the x axis. Find D .



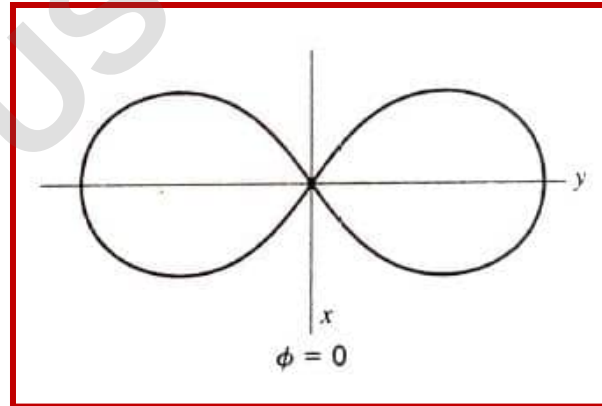
Solution:

The relative (normalized) power pattern in the equatorial plane is

$$P_n = \frac{S_r}{S_{rm}} = \frac{U}{U_m} = \left(\frac{E_\theta}{E_{\theta m}} \right)^2$$

and substituting

$$P_n = \sin^2 \phi$$



Phase Patterns

Phase Patterns

Assuming that the field varies harmonically with time and that the frequency is known, the far field in all directions from a source may be completely specified by a knowledge of the following four quantities:

1. Amplitude of the polar component E_θ of the electric field as a function of r , θ , and ϕ
2. Amplitude of the azimuthal component E_ϕ of the electric field as a function of r , θ , and ϕ
3. Phase lag δ of E_ϕ behind E_θ as a function of θ and ϕ
4. Phase lag η of either field component behind its value at a reference point as a function of r , θ , and ϕ

Figure 5–13 shows the pattern of Fig. 2–3 in three-dimensional, polar and decibel displays. Note that the polarity of the lobes alternate (+ and –). Thus, when the magnitude of the field of one lobe (+) and the adjacent lobe (–) are equal, the total field goes to zero, producing a null.

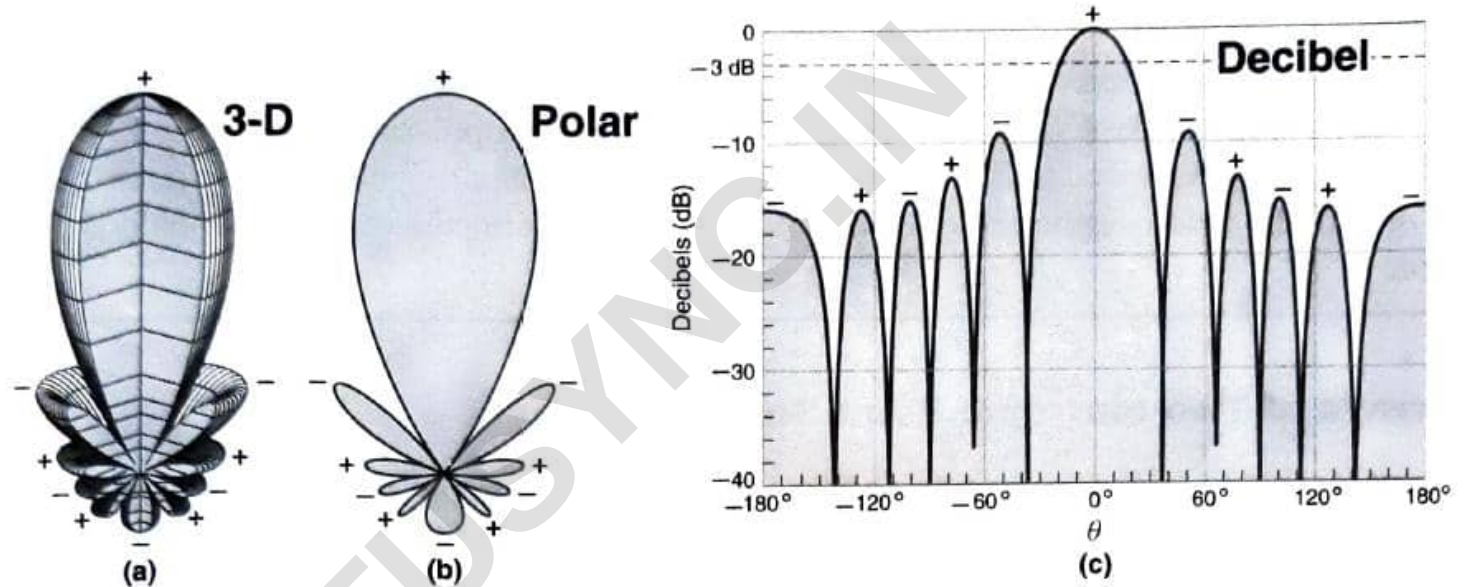


Figure 5-13 Three-dimensional field pattern at (a), polar pattern at (b), and decibel pattern at (c) showing alternate phasing (+ and -) of pattern lobes.

Array of Point Sources

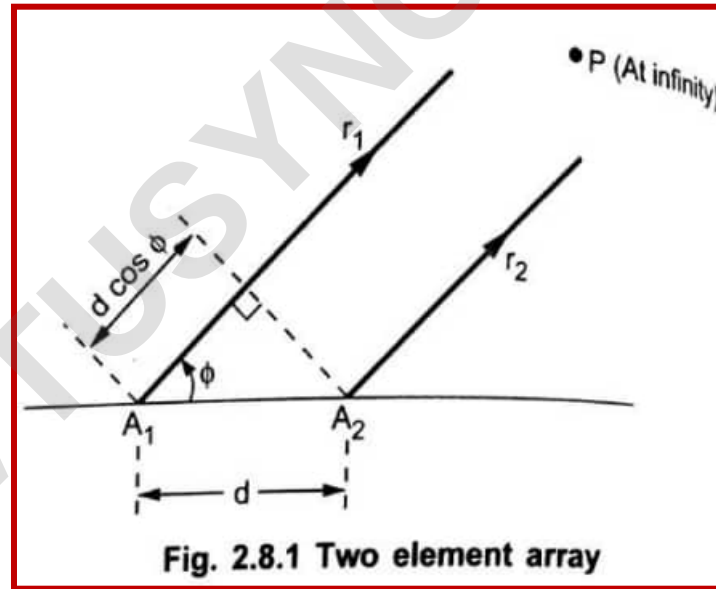
The **array of point sources** is nothing but the array of an isotropic radiators occupying zero volume. For the greater number of point source in the array, the analysis of antenna array becomes complicated and time consuming. Also the simplest condition of number of point sources in the array is two. Then conveniently analysis is done by considering first two point sources, which are separated by distance d and having same polarization. The results obtained for only two point sources can be further extended for 'n' number of point sources in the array.

Let us consider the array of two isotropic point sources, with a distance of separation 'd' between them. The polarization of two isotropic point sources is assumed to be the same. To derive different expressions following conditions can be applied to the antenna array

- i) Two point sources with currents of equal magnitudes and with same phase.
- ii) Two point sources with currents of equal magnitude but with opposite phase.
- iii) Two point sources with currents of unequal magnitudes and with opposite phase.

I. Two Point Sources with Currents Equal in Magnitude and Phase:

Consider two point sources A_1 and A_2 separated by distance d as shown in the Fig. 2.8.1. Consider that both the point sources are supplied with currents equal in magnitude and phase.



Consider point P far away from the array. Let the distance between point P and point sources A_1 and A_2 be r_1 and r_2 respectively. As these radial distances are extremely large as compared with the distance of separation between two point sources i.e. d , we can assume,

$$r_1 = r_2 = r$$

The radiation from the point source A_2 will reach earlier at point P than that from point source A_1 because of the path difference. The extra distance is travelled by the radiated wave from point source A_1 than that by the wave radiated from point source A_2 .

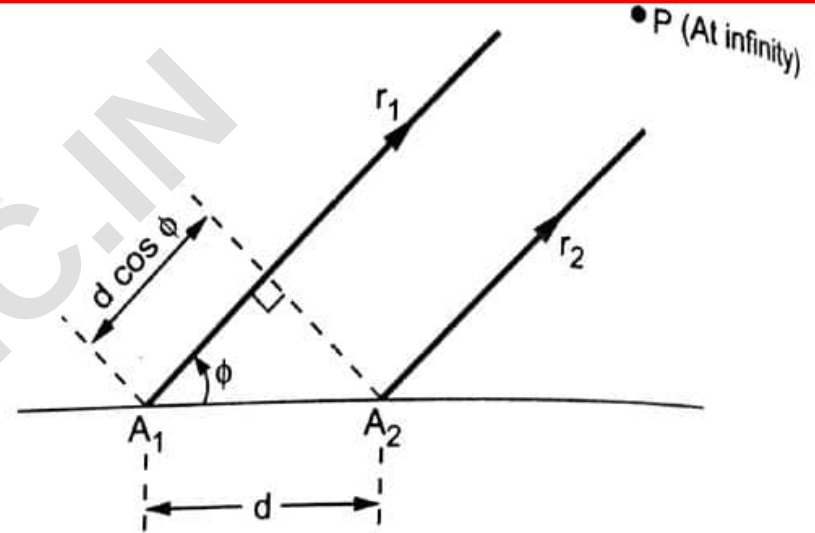


Fig. 2.8.1 Two element array

Hence path difference is given by,

$$\text{Path difference} = d \cos \phi$$

...(2.8.1)

The path difference can be expressed in terms of wavelength as,

$$\text{Path difference} = \frac{d \cos \phi}{\lambda}$$

...(2.8.2)

Hence the phase angle ψ is given by,

$$\text{Phase angle} = \psi = 2\pi (\text{Path difference})$$

$$\psi = 2\pi \left(\frac{d \cos \phi}{\lambda} \right)$$

$$\psi = \frac{2\pi}{\lambda} d \cos \phi \text{ rad}$$

...(2.8.3)

But phase shift $= \beta = \frac{2\pi}{\lambda}$, thus equation (2.8.3) becomes,

$\therefore$

$$\psi = \beta d \cos \phi \text{ rad}$$

...(2.8.4)

Let E_1 be the far field at a distant point P due to point source A_1 . Similarly let E_2 be the far field at point P due to point source A_2 . Then the total field at point P be the addition of the two field components due to the point sources A_1 and A_2 . If the phase angle between the two fields is $\psi = \beta d \cos \phi$ then the far field component at point P due to point source A_1 is given by,

$$E_1 = E_0 \cdot e^{-j\frac{\psi}{2}} \quad \dots(2.8.5)$$

Similarly the far field component at point P due to the point source A_2 is given by,

$$E_2 = E_0 \cdot e^{j\frac{\psi}{2}} \quad \dots(2.8.6)$$

Note that the amplitude of both the field components is E_0 as currents are same and the point sources are identical.

The total field at point P is given by,

$$E_T = E_1 + E_2 = E_0 \cdot e^{-j\frac{\Psi}{2}} + E_0 \cdot e^{j\frac{\Psi}{2}}$$

$$\therefore E_T = E_0 \left(e^{-j\frac{\Psi}{2}} + e^{j\frac{\Psi}{2}} \right)$$

Rearranging the terms on R.H.S., we get,

$$\therefore E_T = 2E_0 \left(\frac{e^{j\frac{\Psi}{2}} + e^{-j\frac{\Psi}{2}}}{2} \right) \quad \text{..(2.8.7)}$$

By trigonometric identity, $\frac{e^{j\theta} + e^{-j\theta}}{2} = \cos \theta$.

Hence equation (2.8.7) can be written as,

$$E_T = 2E_0 \cos\left(\frac{\psi}{2}\right) \quad \dots(2.8.8)$$

Substituting value of ψ from equation (2.8.4), we get,

$$\therefore \quad E_T = 2E_0 \cos\left(\frac{\beta d \cos \phi}{2}\right) \quad \dots(2.8.9)$$

Above equation represents total field intensity at point P, due to two point sources having currents of same amplitude and phase. The total amplitude of the field at point P is $2E_0$ while the phase shift is $\frac{\beta d \cos \phi}{2}$.

- The array factor

The **array factor** is the ratio of the magnitude of the resultant field to the magnitude of the maximum field.

$$\therefore \text{A.F.} = \frac{|E_T|}{|E_{\max}|}$$

But maximum field is $E_{\max} = 2 E_0$.

$$\therefore \text{A.F.} = \frac{|E_T|}{|2 E_0|} = \cos \left(\pi \frac{d}{\lambda} \cos \phi \right)$$

The array factor represents the relative value of the field as a function of ϕ . It defines the radiation pattern in a plane containing the line of the array.

- Maxima direction

From equation (2.8.9), the total field is maximum when $\cos\left(\frac{\beta d \cos \phi}{2}\right)$ is maximum.

As we know, the variation of cosine of a angle is ± 1 . Hence the condition for maxima is given by,

$$\cos\left(\frac{\beta d \cos \phi}{2}\right) = \pm 1$$

Let spacing between the two point sources be $\frac{\lambda}{2}$. Then we can write,

$$\cos\left[\frac{\beta(\lambda/2) \cos \phi}{2}\right] = \pm 1 \quad \dots(2.8.10)$$

i.e. $\cos\left[\frac{\frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} \cos \phi}{2}\right] = \pm 1 \quad \dots \because \beta = \frac{2\pi}{\lambda}$

$$\cos \left[\frac{\beta(\lambda/2) \cos \phi}{2} \right] = \pm 1 \quad \dots(2.8.10)$$

i.e. $\cos \left[\frac{2\pi \cdot \lambda}{\lambda} \cdot \frac{\cos \phi}{2} \right] = \pm 1 \quad \dots \because \beta = \frac{2\pi}{\lambda}$

i.e. $\cos \left(\frac{\pi}{2} \cos \phi \right) = \pm 1$

i.e. $\frac{\pi}{2} \cos \phi_{\max} = \cos^{-1}(\pm 1) = \pm n\pi, \text{ where } n = 0, 1, 2, \dots$

If $n = 0$, then

$$\frac{\pi}{2} \cos \phi_{\max} = 0$$

i.e. $\cos \phi_{\max} = 0$

i.e. $\phi_{\max} = 90^\circ \text{ or } 270^\circ$

$\dots(2.8.11)$

- Minima direction

Again from equation (2.8.9), total field strength is minimum when $\cos\left(\frac{\beta d \cos\phi}{2}\right)$ is minimum i.e. 0 as cosine of angle has minimum value 0. Hence the condition for minima is given by,

$$\therefore \cos\left(\frac{\beta d \cos\phi}{2}\right) = 0 \quad \dots (2.8.12)$$

Again assuming $d = \frac{\lambda}{2}$ and $\beta = \frac{2\pi}{\lambda}$, we can write

$$\cos\left(\frac{\pi}{2} \cos\phi_{\min}\right) = 0$$

$$\therefore \frac{\pi}{2} \cos\phi_{\min} = \cos^{-1} 0 = \pm(2n+1)\frac{\pi}{2}, \text{ where } n = 0, 1, 2, \dots$$

Again assuming $d = \frac{\lambda}{2}$ and $\beta = \frac{2\pi}{\lambda}$, we can write

$$\cos\left(\frac{\pi}{2} \cos\phi_{\min}\right) = 0$$

$$\therefore \frac{\pi}{2} \cos\phi_{\min} = \cos^{-1} 0 = \pm(2n+1)\frac{\pi}{2}, \text{ where } n = 0, 1, 2, \dots$$

If $n = 0$, then,

$$\frac{\pi}{2} \cos\phi_{\min} = \pm \frac{\pi}{2}$$

$$\text{i.e. } \cos\phi_{\min} = \pm 1$$

i.e.

$$\phi_{\min} = 0^\circ \text{ or } 180^\circ$$

...(2.8.13)

- Half power point directions

When the power is half, the voltage or current is $\frac{1}{\sqrt{2}}$ times the maximum value.

Hence the condition for half power point is given by,

$$\cos\left(\frac{\beta d \cos\phi}{2}\right) = \pm \frac{1}{\sqrt{2}} \quad \dots(2.8.14)$$

Let $d = \frac{\lambda}{2}$ and $\beta = \frac{2\pi}{\lambda}$, then we can write,

$$\cos\left(\frac{\pi}{2} \cos\phi\right) = \pm \frac{1}{\sqrt{2}}$$

$$\text{i.e. } \frac{\pi}{2} \cos\phi = \cos^{-1}\left(\pm \frac{1}{\sqrt{2}}\right) = \pm(2n+1)\frac{\pi}{4}, \text{ where } n = 0, 1, 2, \dots$$

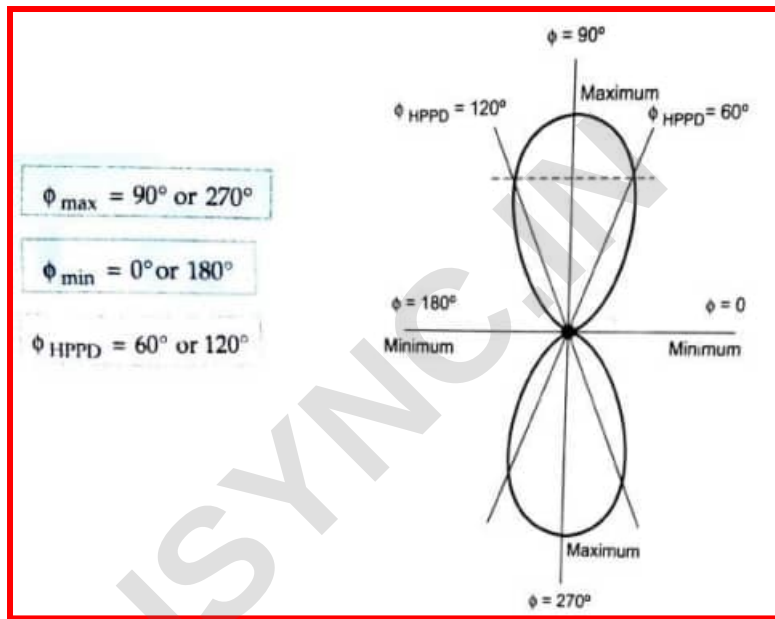
If $n = 0$, then

$$\frac{\pi}{2} \cos\phi_{\text{HPPD}} = \pm \frac{\pi}{4}$$

$$\text{i.e. } \cos\phi_{\text{HPPD}} = \pm \frac{1}{2}$$

$$\text{i.e. } \phi_{\text{HPPD}} = \cos^{-1}\left(\pm \frac{1}{2}\right)$$

$$\therefore \phi_{\text{HPPD}} = 60^\circ \text{ or } 120^\circ \quad \dots(2.8.15)$$

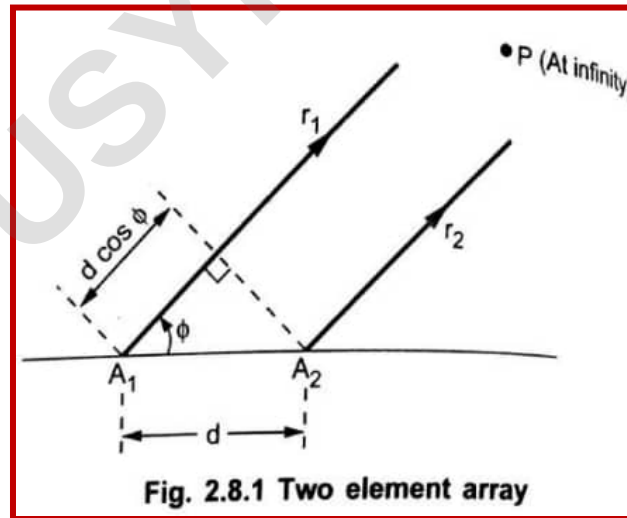


The field pattern drawn with E_T against ϕ for $d = \frac{\lambda}{2}$, then the pattern is bidirectional as shown in the Fig. 2.8.2. The field pattern obtained is bidirectional and it is a figure of eight (8). If this pattern is rotated by 360° about axis, it will represent three dimensional doughnut shaped space pattern. This is the simplest type of broadside array of two point sources and it is called **Broadside couplet** as two radiations of point sources are in phase.

2. Two Point Sources with Currents Equal in Magnitude and Opposite in Phase:

Consider two point sources separated by distance d and supplied with currents equal in magnitude but opposite in phase. Consider Fig. 2.8.1 from section 2.8. All the conditions are exactly same except the phase of the currents is opposite i.e. 180° . With this condition, the total field at far point P is given by,

$$E_T = (-E_1) + (E_2) \quad \dots(2.9.1)$$



Assuming equal magnitudes of currents, the fields at point P due to the point sources A_1 and A_2 can be written as,

$$E_1 = E_0 e^{-j\frac{\psi}{2}} \quad \dots(2.9.2)$$

and $E_2 = E_0 e^{j\frac{\psi}{2}} \quad \dots(2.9.3)$

Substituting values of E_1 and E_2 in equation (2.9.1), we get,

$$E_T = -E_0 \cdot e^{-j\frac{\psi}{2}} + E_0 \cdot e^{j\frac{\psi}{2}}$$

$$\therefore E_T = E_0 \left(-e^{-j\frac{\psi}{2}} + e^{j\frac{\psi}{2}} \right)$$

Rearranging the terms in above equation, we get,

$$\therefore E_T = (j2) E_0 \left(\frac{e^{j\frac{\psi}{2}} - e^{-j\frac{\psi}{2}}}{j2} \right) \quad \dots(2.9.4)$$

By trigonometric identity, $\frac{e^{j\theta} - e^{-j\theta}}{2} = \sin \frac{\theta}{2}$.

Hence equation (2.9.4) can be written as,

$$E_T = j2 E_0 \sin\left(\frac{\Psi}{2}\right) \quad \dots(2.9.5)$$

Now as the condition for two point sources with currents in phase and out of phase is exactly same, the phase angle can be written as previous case,

$$\text{Phase angle} = \psi = \beta d \cos \phi \quad \dots(2.9.6)$$

Substituting value of phase angle in equation (2.9.5), we get,

$$E_T = j(2E_0) \sin\left(\frac{\beta d \cos \phi}{2}\right) \quad \dots(2.9.7)$$

- Maxima direction

From equation (2.9.7), the total field is maximum when $\sin\left(\frac{\beta d \cos\phi}{2}\right)$ is maximum i.e. ± 1 as the maximum value of sine of angle is ± 1 . Hence condition for maxima is given by,

$$\sin\left(\frac{\beta d \cos\phi}{2}\right) = \pm 1$$

Let the spacing between two isotropic point sources be equal to $\frac{\lambda}{2}$. i.e. $d = \frac{\lambda}{2}$.

Substituting $d = \frac{\lambda}{2}$ and $\beta = \frac{2\pi}{\lambda}$ in equation (2.9.8), we get,

$$\sin\left(\frac{\pi}{2} \cos\phi\right) = \pm 1$$

$$\text{i.e. } \frac{\pi}{2} \cos\phi = \pm(2n+1)\frac{\pi}{2}, \text{ where } n = 0, 1, 2, \dots$$

If $n = 0$, then

$$\frac{\pi}{2} \cos\phi_{\max} = \pm \frac{\pi}{2}$$

$$\text{i.e. } \cos\phi_{\max} = \pm 1$$

$$\text{i.e. } \phi_{\max} = 0^\circ \text{ and } 180^\circ$$

...(2.9.9)

- Minima direction

Again from equation (2.9.7), total field strength is minimum when $\sin\left(\frac{\beta d \cos\phi}{2}\right)$ is minimum i.e. 0.

Hence the condition for minima is given by,

$$\sin\left(\frac{\beta d \cos\phi}{2}\right) = 0 \quad \dots(2.9.10)$$

Assuming $d = \frac{\lambda}{2}$ and $\beta = \frac{2\pi}{\lambda}$ in equation (2.9.10), we get,

$$\sin\left(\frac{\pi}{2} \cos\phi\right) = 0$$

$$\text{i.e. } \frac{\pi}{2} \cos\phi = \pm n\pi, \text{ where } n = 0, 1, 2, \dots$$

If $n = 0$, then

$$\frac{\pi}{2} \cos\phi_{\min} = 0$$

$$\text{i.e. } \cos\phi_{\min} = 0$$

$$\text{i.e. } \phi_{\min} = +90^\circ \text{ or } -90^\circ$$

...(2.9.11)

- Half power point directions

When the power is half of maximum value, the voltage or current equals to $\frac{1}{\sqrt{2}}$ times the respective maximum value. Hence the condition for the half power point can be obtained from equation (2.9.7) as,

$$\sin\left(\frac{\beta d \cos\phi}{2}\right) = \pm \frac{1}{\sqrt{2}} \quad \dots(2.9.12)$$

Let $d = \frac{\lambda}{2}$ and $\beta = \frac{2\pi}{\lambda}$, we can write,

$$\sin\left(\frac{\pi}{2} \cos\phi\right) = \pm \frac{1}{\sqrt{2}}$$

i.e. $\frac{\pi}{2} \cos\phi = \pm(2n+1)\frac{\pi}{4}$, where $n = 0, 1, 2$.

If $n = 0$, we can write,

$$\frac{\pi}{2} \cos\phi_{\text{HPPD}} = \pm \frac{\pi}{4}$$

i.e. $\cos\phi_{\text{HPPD}} = \pm \frac{1}{2}$

$$\therefore \phi_{\text{HPPD}} = 60^\circ \text{ or } 120^\circ \quad \dots(2.9.13)$$

Thus from the conditions of maxima, minima and half power points, the field pattern can be drawn as shown in the Fig. 2.9.1.

As compared with the field pattern for two point sources with in-phase currents, the maxima have shifted by 90° along X-axis in case of out-phase currents in two point source array. Thus the maxima is along the axis of the array or along the line joining two point sources. In first case, we have obtained vertical figure of 8. Now in

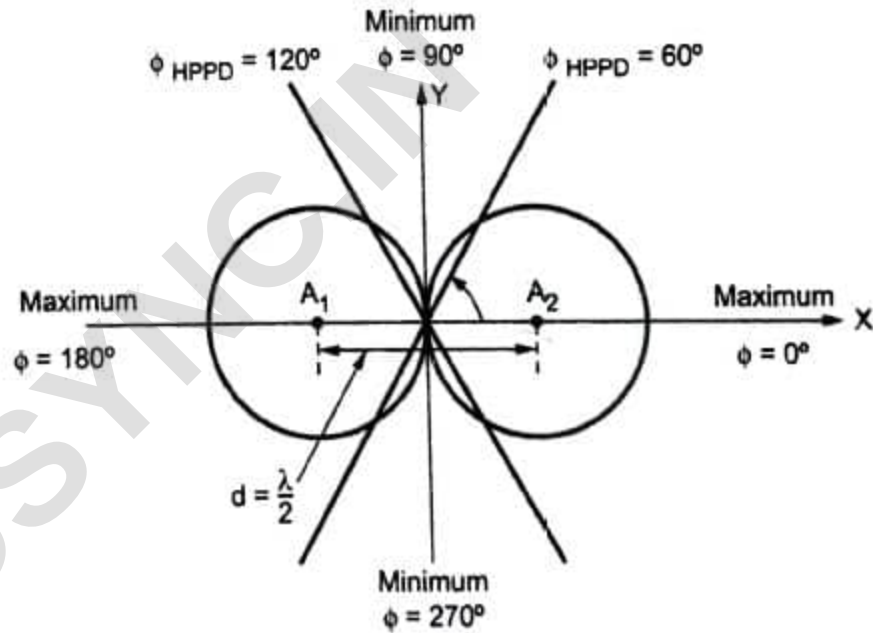


Fig. 2.9.1 Field pattern for two point sources with spacing $d = \frac{\lambda}{2}$ and fed with currents equal in magnitude but out of phase by 180°

3. Two Point Sources with Currents Unequal in Magnitude and with any Phase:

Let us consider Fig. 2.10.1 from section 2.8. Assume that the two point sources are separated by distance d and supplied with currents which are different in magnitudes and with any phase difference say α . Consider that source 1 is assumed to be reference for phase and amplitude of the fields E_1 and E_2 , which are due to source 1 and source 2 respectively at the distant point P. Let us assume that E_1 is greater than E_2 in magnitude as shown in the vector diagram in Fig. 2.10.1.

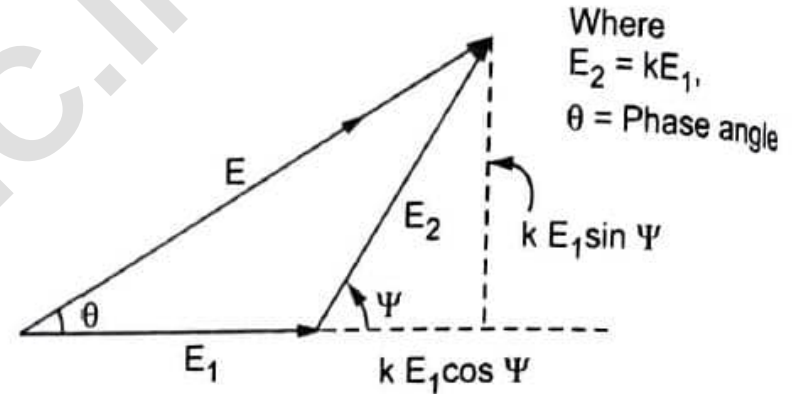


Fig. 2.10.1 Vector diagram of fields E_1 and E_2

Now the total phase difference between the radiations by the two point sources at any far point P is given by,

$$\Psi = \frac{2\pi}{\lambda} \cos\phi + \alpha \quad \dots (2.10.1)$$

where α is the phase angle with which current I_2 leads current I_1 . Now if $\alpha = 0$, then the condition is similar to the **two point sources with currents equal in magnitude and phase**. Similarly if $\alpha = 180^\circ$, then the condition is similar to the **two point source with currents equal in magnitude but opposite in phase**. Assume value of phase difference α as $0 < \alpha < 180^\circ$. Then the resultant field at point P is given by,

$$E_T = E_1 e^{j0} + E_2 e^{j\psi} \quad \dots (\text{source 1 is assumed to be}$$

$$\therefore E_T = E_1 + E_2 e^{j\psi} \quad \text{reference hence phase angle is } 0)$$

$$\therefore E_T = E_1 \left(1 + \frac{E_2}{E_1} e^{j\psi} \right)$$

Let $\frac{E_2}{E_1} = k$... (2.10.2)

Note that $E_1 > E_2$, the value of k is less than unity. Moreover the value of k is given by,
 $0 \leq k \leq 1$.

$$E_T = E_1 [1 + k (\cos \psi + j \sin \psi)] \quad \dots (2.10.3)$$

The magnitude of the resultant field at point P is given by,

$$|E_T| = |E_1 [1 + k \cos \psi + j k \sin \psi]|$$

$$|E_T| = E_1 \sqrt{(1 + k \cos \psi)^2 + (k \sin \psi)^2} \quad \dots (2.10.4)$$

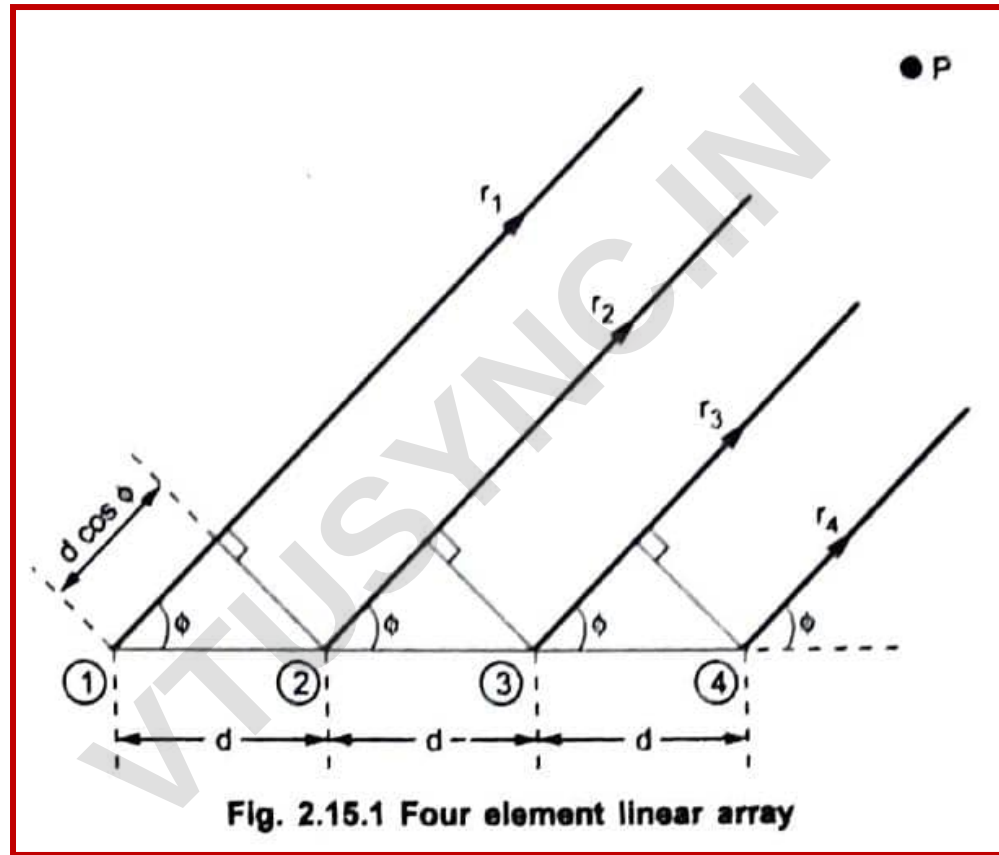
The phase angle between two fields at the far point P is given by,

$$\theta = \tan^{-1} \frac{k \sin \psi}{1 + k \cos \psi} \quad \dots (2.10.5)$$

Pattern Multiplication Method

In the previous sections, we have discussed antenna arrays and the methods of obtaining the radiation patterns of the same. Other than this straight forward mathematical method, there is another simple method of obtaining the same patterns of the arrays. This method is known as **pattern multiplication method**. This method is a very useful in the design of arrays because it makes possible to draw the patterns of complicated arrays rapidly, almost by inspection.

To illustrate this method, consider 4 element array of equispaced identical antennas as shown in the Fig. 2.15.1. Let the spacing between two units be $d = \frac{\lambda}{2}$. Also assume that all the elements are supplied with equal magnitude currents which are in phase.



As the point P at which the resultant field has to be obtained is far away, we can assume the radiation from the antenna in the form of parallel lines.

The radiation pattern of the antennas (1) and (2) treated to be operating as a single unit is as shown in the Fig. 2.15.2 (a). Similarly the radiation pattern of the antennas (3)

and (4), spaced $\frac{\lambda}{2}$ distance apart and fed with equal current in phase, treated to be operated as single unit is again as shown in the Fig. 2.15.2 (a). Now instead of considering two separate elements (1) and (2), we can replace it by a single antenna located at a point midway between them $\left(\frac{d}{2}\right)$ as shown in the Fig. 2.15.2 (c).

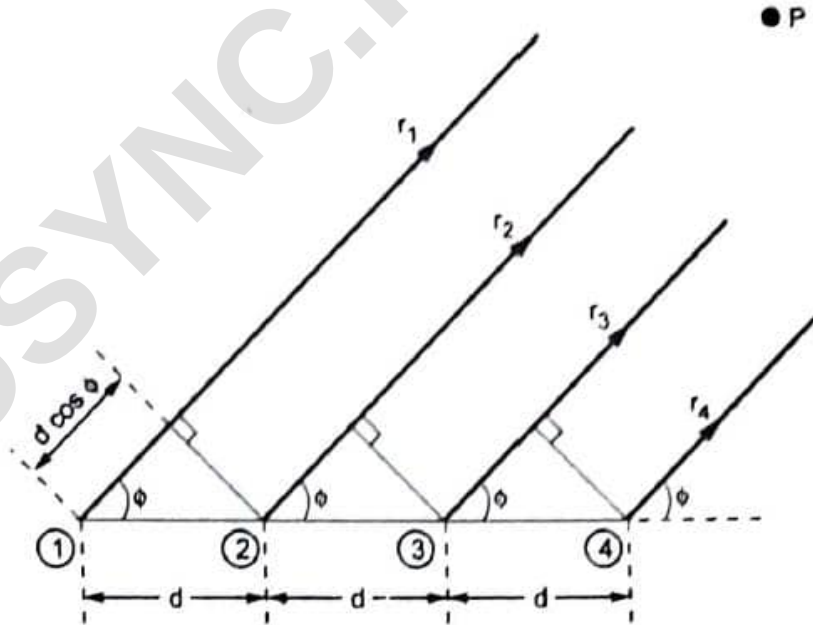
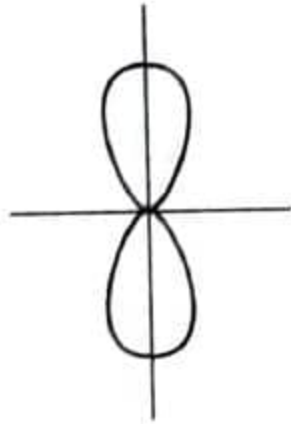
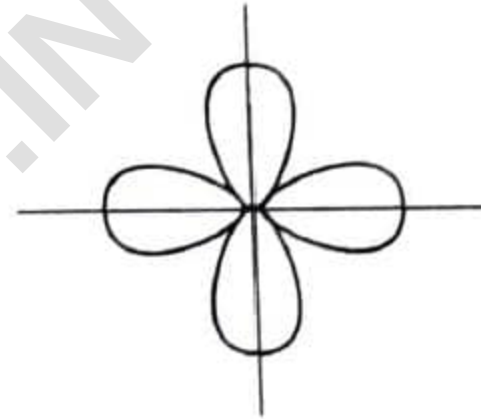


Fig. 2.15.1 Four element linear array

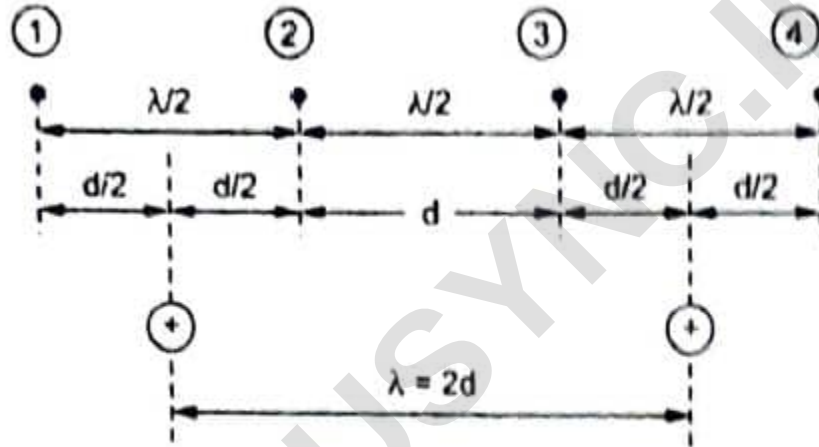
in the Fig. 2.15.2 (c). Similarly replacing antennas (3) and (4) by single antenna having same pattern as shown in the Fig. 2.15.2 (c). Now both the antennas have bidirection pattern i.e. figure eight pattern spaced distance λ apart from each other, fed with equal currents in phase is as shown in the Fig. 2.15.2 (b). Now the resultant radiation pattern of four element array can be obtained as the multiplication of pattern as shown in the Fig. 2.15.2 (d). Note that this multiplication is polar graphical multiplication for different values of ϕ .



(a) Radiation pattern of two antennas spaced at distance $\frac{\lambda}{2}$ and fed with equal currents in phase

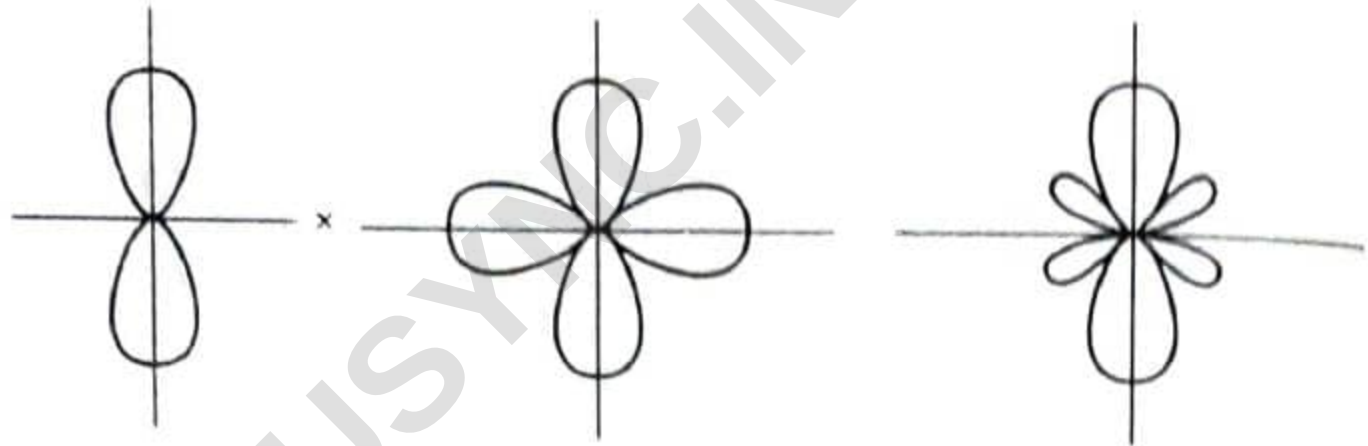


(b) Radiation pattern of two antennas spaced at distance λ and fed with equal currents in phase



Array of 4 identical elements,
Replacement of array by two single
antennas placed at distance λ apart

(c) Antenna ① and ② and ③ and ④ replaced
by single antenna separately



(d) Multiplication of pattern

Fig. 2.15.2 Illustration of pattern multiplication method

The field pattern of an array of nonisotropic but similar point sources is the product of the pattern of the individual source and the pattern of an array of isotropic point sources having the same locations, relative amplitudes, and phase as the nonisotropic point sources.

The total phase pattern is referred to the phase center of the array. In symbols, the total field E is then

$$E = \underbrace{f(\theta, \phi)}_{\text{Field pattern}} \underbrace{F(\theta, \phi)}_{\text{Phase pattern}} \quad (4)$$

where

- $f(\theta, \phi)$ = field pattern of individual source
- $f_p(\theta, \phi)$ = phase pattern of individual source
- $F(\theta, \phi)$ = field pattern of array of isotropic sources
- $F_p(\theta, \phi)$ = phase pattern of array of isotropic sources

The patterns are expressed in (4) as a function of both polar angles to indicate that the principle of pattern multiplication applies to space patterns as well as to the two-dimensional cases we have been considering.

n-Element Uniform Linear Arrays

At higher frequencies, for point to point communications it is necessary to have a pattern with single beam radiation. Such highly directive single beam pattern can be obtained by increasing the point sources in the array from 2 to n say.

An array of n elements is said to be **linear array** if all the individual elements are spaced equally along a line. An array is said to be **uniform array** if the elements in the array are fed with currents with equal magnitudes and with uniform progressive phase shift along the line.

Consider a general n element linear and uniform array with all the individual elements spaced equally at distance d from each other and all elements are fed with currents equal in magnitude and uniform progressive phase shift along line as shown in the Fig. 2.11.1. (See Fig. 2.11.1 on next page)

The total resultant field at the distant point P is obtained by adding the fields due to n individual sources vectorically. Hence we can write,

$$E_T = E_0 \cdot e^{j0} + E_0 e^{j\psi} + E_0 e^{2j\psi} + \dots + E_0 e^{j(n-1)\psi}$$

$$\therefore E_T = E_0 [1 + e^{j\psi} + e^{j2\psi} + \dots + e^{j(n-1)\psi}] \quad \dots (2.11.1)$$

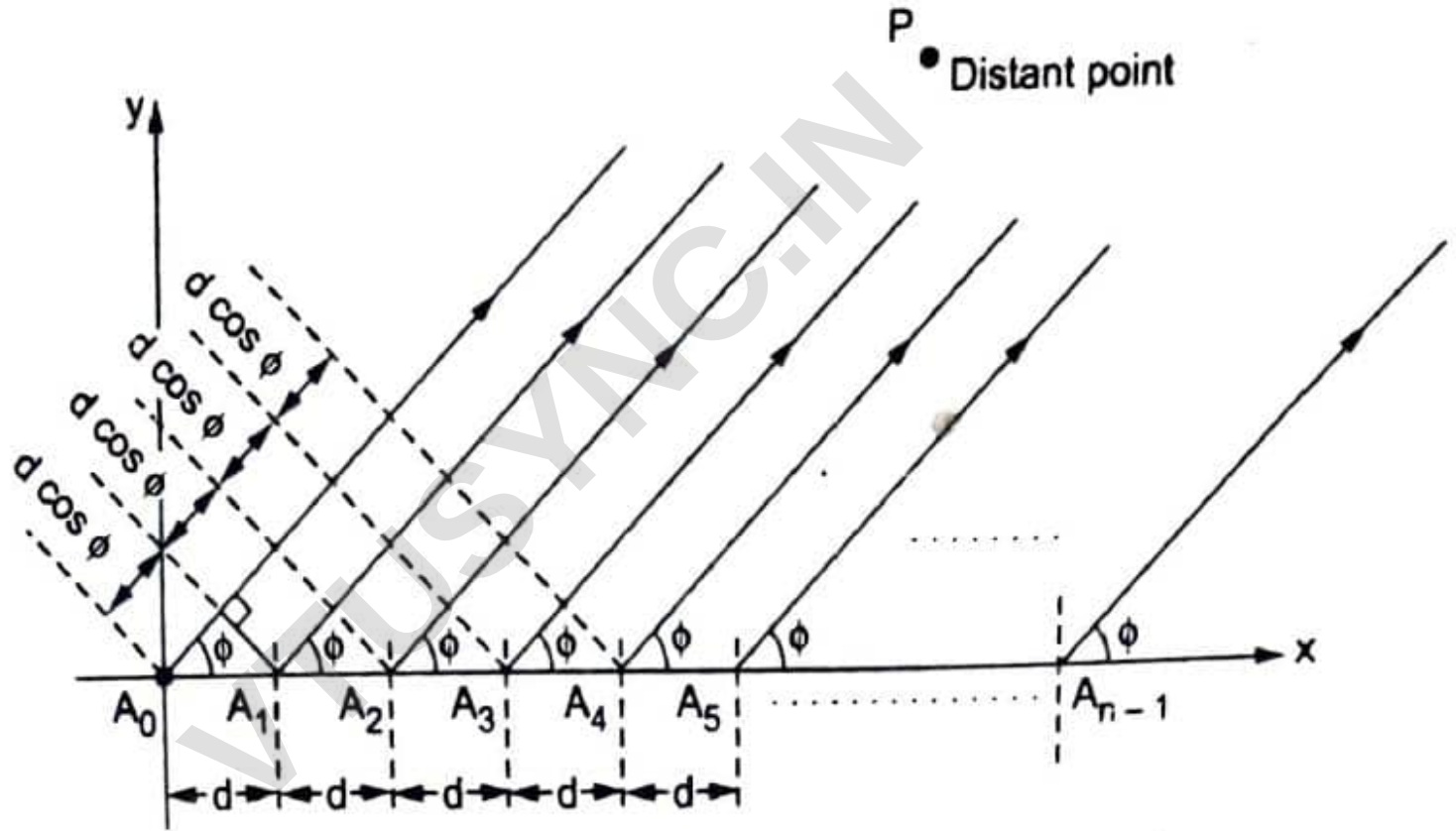


Fig. 2.11.1 Uniform, linear array of n elements

Note that $\psi = (\beta d \cos \phi + \alpha)$ indicates the total phase difference of the fields from adjacent sources calculated at point P. Similarly α is the progressive phase shift between two adjacent point sources. The value of α may lie between 0° and 180° . If $\alpha = 0^\circ$, we get **n element uniform linear broadside array**. If $\alpha = 180^\circ$, we get **n element uniform linear end fire array**.

Multiplying equation (2.11.1) by $e^{j\psi}$, we get,

$$E_T e^{j\psi} = E_0 [e^{j\psi} + e^{j2\psi} + e^{j3\psi} + \dots + e^{jn\psi}] \quad \dots(2.11.2)$$

Subtracting equation (2.11.2) from (2.11.1), we get,

$$E_T - E_T e^{j\psi} = E_0 \{ [1 + e^{j\psi} + e^{j2\psi} + \dots + e^{j(n-1)\psi}] - [e^{j\psi} + e^{j2\psi} + \dots + e^{jn\psi}] \}$$

$$E_T (1 - e^{j\psi}) = E_0 (1 - e^{jn\psi})$$

$$\therefore E_T = E_0 \left[\frac{1 - e^{jn\psi}}{1 - e^{j\psi}} \right] \quad \dots (2.11.3)$$

Simplifying mathematically, we get

$$E_T = E_0 \left[\frac{e^{j\frac{n\psi}{2}} \left(e^{-j\frac{n\psi}{2}} - e^{j\frac{n\psi}{2}} \right)}{e^{j\frac{\psi}{2}} \left(e^{-j\frac{\psi}{2}} - e^{j\frac{\psi}{2}} \right)} \right]$$

According to trigonometric identity,

$$e^{-j\theta} - e^{j\theta} = -2j \sin \theta,$$

The resultant field is given by,

$$E_T = E_0 \left[\frac{\left(-j2\sin \frac{n\psi}{2} \right) e^{j\frac{n\psi}{2}}}{\left(-j2\sin \frac{\psi}{2} \right) e^{j\frac{\psi}{2}}} \right]$$

∴

$$E_T = E_0 \left[\frac{\sin n \frac{\psi}{2}}{\sin \frac{\psi}{2}} \right] e^{j \left(\frac{n-1}{2} \right) \psi}$$

... (2.11.4)

This equation (2.11.4) indicates the resultant field due to n element array at distant point P.

The magnitude of the resultant field is given by,

∴

$$E_T = E_0 \left[\frac{\sin n \frac{\psi}{2}}{\sin \frac{\psi}{2}} \right]$$

... (2.11.5)

The phase angle θ of the resultant field at point P is given by,

∴

$$\theta = \frac{(n-1)}{2} \psi = \frac{(n-1)}{2} \beta d \cos \phi + \alpha$$

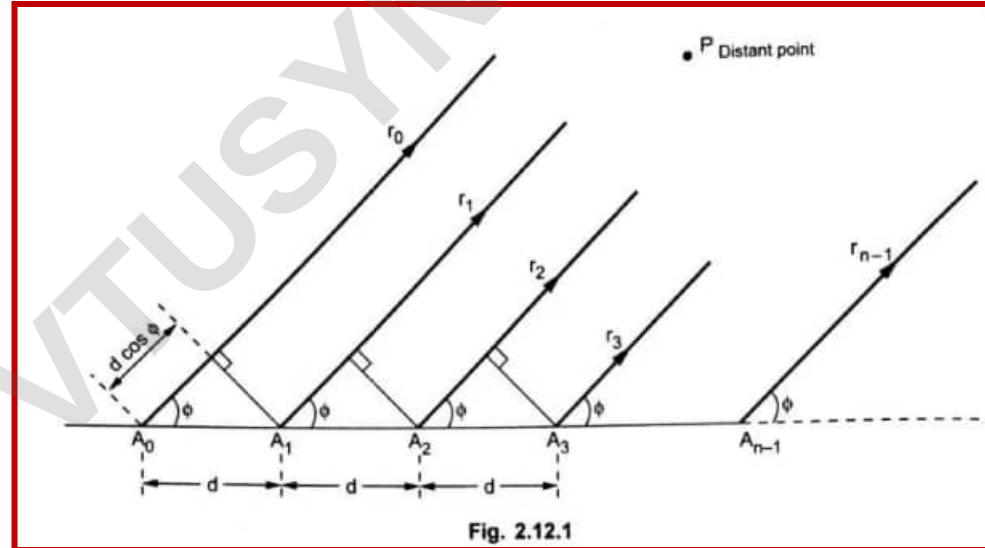
... (2.11.6)

Array of n Elements with Equal Spacing and Currents Equal in Magnitude and Phase - Broadside Array

Array of n Elements with Equal Spacing and Currents Equal in Magnitude and Phase

Consider the ' n ' number of identical radiators carry currents which are equal in magnitude and in phase. The identical radiators are equispaced. Hence the maximum radiation occurs in the directions normal to the line of array. Hence such an array is known as **Uniform broadside array**.

Consider a broadside array with n identical radiators as shown in the Fig. 2.12.1.



The electric field produced at point P due to an element A_0 is given by,

$$E_0 = \frac{I dL \sin \theta}{4 \pi \omega \epsilon_0} \left[j \frac{\beta^2}{r_0} \right] e^{-j\beta r_0} \quad \dots (2.12.1)$$

As the distance of separation d between any two array elements is very small as compared to the radial distances of point P from $A_0, A_1, \dots, A_{n-1}$, we can assume $r_0, r_1, r_2, \dots, r_{n-1}$ are approximately same.

Now the electric field produced at point P due to an element A_1 will differ in phase as r_0 and r_1 are not actually same. Hence the electric field due to A_1 is given by,

$$E_1 = \frac{I dL \sin \theta}{4 \pi \omega \epsilon_0} \left[j \frac{\beta^2}{r_1} \right] e^{-j\beta r_1}$$

But

$$r_1 = r_0 - d \cos \phi$$

$$\therefore E_1 = \frac{I dL \sin \theta}{4 \pi \omega \epsilon_0} \left[j \frac{\beta^2}{r_0} \right] e^{-j\beta(r_0 - d \cos \phi)} \quad \dots r_1 \approx r_0$$

$$\therefore E_1 = \frac{I dL \sin \theta}{4 \pi \omega \epsilon_0} \left[j \frac{\beta^2}{r_0} \right] e^{-j\beta r_0} e^{j\beta d \cos \phi}$$

$$\therefore E_1 = E_0 \cdot e^{j\beta d \cos \phi} \quad \dots (2.12.2)$$

Exactly on the similar lines we can write the electric field produced at point P due to an element A_2 as,

$$E_2 = \frac{I dL \sin \theta}{4 \pi \omega \epsilon_0} \left[j \frac{\beta^2}{r_2} \right] e^{-j\beta r_2}$$

$$\therefore E_2 = \frac{I dL \sin \theta}{4 \pi \omega \epsilon_0} \left[j \frac{\beta^2}{r_1} \right] e^{-j\beta(r_1 - d \cos \phi)} \quad \dots r_2 = r_1 - d \cos \phi$$

$$\therefore E_2 = \left\{ \frac{I dL \sin \theta}{4 \pi \omega \epsilon_0} \left[j \frac{\beta^2}{r_1} \right] e^{-j\beta r_1} \right\} e^{j\beta d \cos \phi}$$

But the term inside the bracket represents E_1

$$\therefore E_2 = E_1 e^{j\beta d \cos \phi}$$

From equation (2.12.2), substituting value of E_1 , we get,

$$E_2 = \left[E_0 e^{j\beta d \cos \phi} \right] e^{j\beta d \cos \phi}$$

$$\therefore E_2 = E_0 \cdot e^{j2\beta d \cos \phi} \quad \dots (2.12.3)$$

Similarly the electric field produced at point P due to an element A_{n-1} is given by,

$$E_{n-1} = E_0 \cdot e^{j(n-1)\beta d \cos \phi} \quad \dots (2.12.4)$$

The total electric field at point P is given by,

$$E_T = E_0 + E_1 + E_2 + \dots + E_{n-1}$$

$$\therefore E_T = E_0 + E_0 e^{j\beta d \cos \phi} + E_0 e^{j2\beta d \cos \phi} + \dots + E_0 e^{j(n-1)\beta d \cos \phi}$$

Let $\beta d \cos \phi = \psi$, then rewriting above equation,

$$E_T = E_0 + E_0 e^{j\psi} + E_0 e^{j2\psi} + \dots + E_0 e^{j(n-1)\psi}$$

$$\therefore E_T = E_0 [1 + e^{j\psi} + e^{j2\psi} + \dots + e^{j(n-1)\psi}] \quad \dots (2.12.5)$$

$$\therefore E_T = E_0 [1 + e^{j\psi} + e^{j2\psi} + \dots + e^{j(n-1)\psi}] \quad \dots (2.12.5)$$

Consider a series given by

$$s = 1 + r + r^2 + r^3 + \dots + r^{n-1} \quad \dots (i)$$

where $r = e^{j\psi}$

Multiplying both the sides of the equation (i) by r ,

$$s.r = r + r^2 + r^3 + r^4 + \dots + r^n \quad \dots (ii)$$

Subtracting equation (ii) from (i), we get,

$$s - sr = 1 - r^n$$

$$\therefore s(1 - r) = 1 - r^n$$

$$\therefore s = \frac{1 - r^n}{1 - r} \quad \dots (iii)$$

Using equation (iii), equation (2.12.5) can be modified as,

$$E_T = E_0 \left[\frac{1 - e^{jn\psi}}{1 - e^{j\psi}} \right]$$

$$E_T = E_0 \left[\frac{1 - e^{jn\psi}}{1 - e^{j\psi}} \right]$$

$$\therefore \frac{E_T}{E_0} = \frac{e^{jn\frac{\psi}{2}} \left[e^{-jn\frac{\psi}{2}} - e^{jn\frac{\psi}{2}} \right]}{e^{j\frac{\psi}{2}} \left[e^{-j\frac{\psi}{2}} - e^{j\frac{\psi}{2}} \right]} \quad \dots (2.12.6)$$

From the trigonometric identities

$$e^{-j\theta} = \cos \theta - j \sin \theta$$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$\text{and } e^{-j\theta} - e^{j\theta} = -j 2 \sin \theta$$

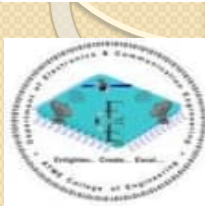
Using above trigonometric identities, equation (2.12.6) can be written as,

$$\frac{E_T}{E_0} = \frac{e^{jn\frac{\psi}{2}} \left[-j 2 \sin\left(\frac{n\psi}{2}\right) \right]}{e^{j\frac{\psi}{2}} \left[-j 2 \sin\left(\frac{\psi}{2}\right) \right]}$$

$$\therefore \frac{E_T}{E_0} = e^{j\frac{(n-1)\psi}{2}} \left[\frac{\sin\left(\frac{n\psi}{2}\right)}{\sin\left(\frac{\psi}{2}\right)} \right] \quad \dots (2.12.7)$$

The exponential term in equation (2.12.7) represents the phase shift. Now considering magnitudes of the electric fields, we can write,

$$\left| \frac{E_T}{E_0} \right| = \frac{\sin \frac{n\psi}{2}}{\sin \frac{\psi}{2}} \quad \dots (2.12.8)$$



Numeric Problems

Table: Summary of n element uniform linear broadside array

Sr. No.	Parameter	Expressions
1.	Direction of Major Lobe	$\phi_{\max} = 90^\circ$ or $\phi_{\max} = 270^\circ$
2.	Magnitude of Major Lobe	n
3.	Direction of Minor Lobe	$\phi_{\min} = \cos^{-1} \left[\pm \frac{m\lambda}{nd} \right]$ where m = 1, 2, 3, ...
4.	Direction of Subsidiary Maxima (or Side Lobes)	$\phi = \cos^{-1} \left[\pm \frac{\lambda(2m+1)}{2nd} \right]$ where m = 1, 2, 3, ...
5.	Beamwidth of Major Lobe (BWFN)	$BWFN = \frac{2\lambda}{L} = \frac{2}{\left(\frac{L}{\lambda}\right)} \text{ rad} = \frac{114.6}{\left(\frac{L}{\lambda}\right)} \text{ degree}$
6.	Half Power Beam Width (HPBW)	$HPBW = \frac{BWFN}{2} = \frac{1}{\left(\frac{L}{\lambda}\right)} \text{ rad} = \frac{57.3}{\left(\frac{L}{\lambda}\right)} \text{ degree}$
7.	Directivity	$D = 2 \left(\frac{nd}{\lambda} \right) = 2 \left(\frac{L}{\lambda} \right)$

Example Using exact method, calculate the distance between elements of broadside array whose beam width between first null is found to be 45° at a frequency of 10 MHz. There are 8 elements in the array.

Solution : Given : $n = 8$, BWFN = 45° $f = 10 \text{ MHz} = 10 \times 10^6 \text{ Hz}$

Now $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10 \times 10^6} = 30 \text{ m}$

For Broadside array,

$$\text{BWFN} = \frac{114.6}{\left(\frac{L}{\lambda}\right)}$$

But $L = nd$

$$\therefore \text{BWFN} = \frac{114.6}{\left(\frac{nd}{\lambda}\right)} = 45$$

$$\therefore d = \frac{(114.6)\lambda}{45n} = \frac{(114.6)(30)}{45(8)} = 9.55 \text{ m or } 0.31833 \lambda$$

Example 2.12.3 Draw the polar diagram of a broadside array with number of elements = 5 and spacing = $\lambda/2$

Solution : Given : $n = 5$, $d = \lambda/2$

1) Major lobe : For broadside array, the direction of maxima is along direction normal to axis of array. Hence it is given by,

$$\phi = 90^\circ \quad \text{and} \quad \phi = 270^\circ$$

2) Magnitude of major lobe : $|\text{Major lobe}| = n = 5$

3) Nulls : The direction of nulls is given by,

$$\phi_{\min} = \cos^{-1} \left[\pm \frac{m\lambda}{nd} \right] \quad \text{where } m = 1, 2, 3, 4, \dots$$

3) Nulls : The direction of nulls is given by,

$$\phi_{\min} = \cos^{-1} \left[\pm \frac{m\lambda}{nd} \right] \text{ where } m = 1, 2, 3, 4, \dots$$

$$\text{For } m = 1, \phi_{\min 1} = \cos^{-1} \left[\pm \frac{(1)\lambda}{(5)(\lambda/2)} \right] = \cos^{-1} \left[\pm \frac{2}{5} \right] = 66.42^\circ \text{ and } 113.58^\circ$$

$$\text{For } m = 2, \phi_{\min 2} = \cos^{-1} \left[\pm \frac{(2)\lambda}{(5)(\lambda/2)} \right] = \cos^{-1} \left[\pm \frac{4}{5} \right] = 36.87^\circ \text{ and } 143.13^\circ$$

4) Subsidiary maxima (Side Lobes) : The directions of side lobes is given by,

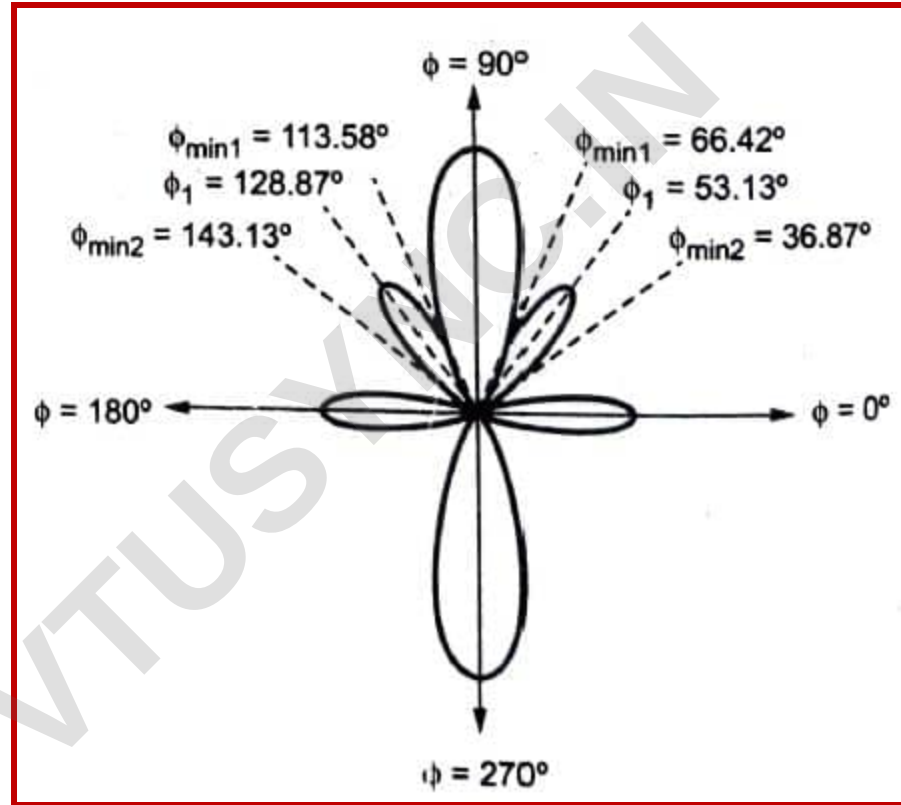
$$\phi = \cos^{-1} \left[\pm \frac{\lambda (2m+1)}{2nd} \right] \text{ where } m = 1, 2, 3, \dots$$

4) Subsidiary maxima (Side Lobes) : The directions of side lobes is given by,

$$\phi = \cos^{-1} \left[\pm \frac{\lambda (2m+1)}{2nd} \right] \text{ where } m = 1, 2, 3, \dots$$

$$\text{For } m = 1, \phi_1 = \cos^{-1} \left[\pm \frac{\lambda(2+1)}{2(5)(\lambda/2)} \right] = \cos^{-1} \left[\pm \frac{3}{5} \right] = 53.13^\circ \text{ and } 128.87^\circ$$

$$\text{For } m = 2, \phi_2 = \cos^{-1} \left[\pm \frac{\lambda(4+1)}{2(5)(\lambda/2)} \right] = \cos^{-1} \left[\pm \frac{5}{5} \right] = 0^\circ \text{ and } 180^\circ$$



Example 2.12.4 Obtain the field pattern for a linear uniform array of 6 isotropic point sources spaced $\lambda / 2$ distance apart. The power is applied with equal amplitude and in phase. Also find HPBW and FNBW.

Solution : Given : $n = 6$, $d = \frac{\lambda}{2}$

i) **Field pattern :**

1) **Major lobe :** For broadside array (equal amplitude and phase) the maxima is along direction normal to axis of array. Hence direction of major lobe is given by $\phi = 90^\circ$ and $\phi = 270^\circ$.

2) **Magnitude of major lobe :** The magnitude of major lobe is given by,

$$|\text{major lobe}| = n = 6$$

3) **Nulls :** The directions of nulls are given by,

$$\phi_{\min} = \cos^{-1} \left[\pm \frac{m\lambda}{nd} \right] \text{ where } m = 1, 2, 3, \dots$$

3) Nulls : The directions of nulls are given by,

$$\phi_{\min} = \cos^{-1} \left[\pm \frac{m\lambda}{n d} \right] \text{ where } m = 1, 2, 3, \dots$$

$$\text{For } m = 1, \phi_{\min 1} = \cos^{-1} \left[\pm \frac{(1)\lambda}{\left(\frac{\lambda}{2}\right)(6)} \right] = \cos^{-1} \left[\pm \frac{1}{3} \right] = 70.53^\circ \text{ and } 109.47^\circ$$

$$\text{For } m = 2, \phi_{\min 2} = \cos^{-1} \left[\pm \frac{(2)\lambda}{\left(\frac{\lambda}{2}\right)^6} \right] = \cos^{-1} \left[\pm \frac{2}{3} \right] = 48.18^\circ \text{ and } 138.81^\circ$$

4) Subsidiary maxima (side lobes) : The direction of side lobes is given by,

$$\phi = \cos^{-1} \left[\pm \frac{\lambda (2m+1)}{2nd} \right] \text{ where } m = 1, 2, 3, \dots$$

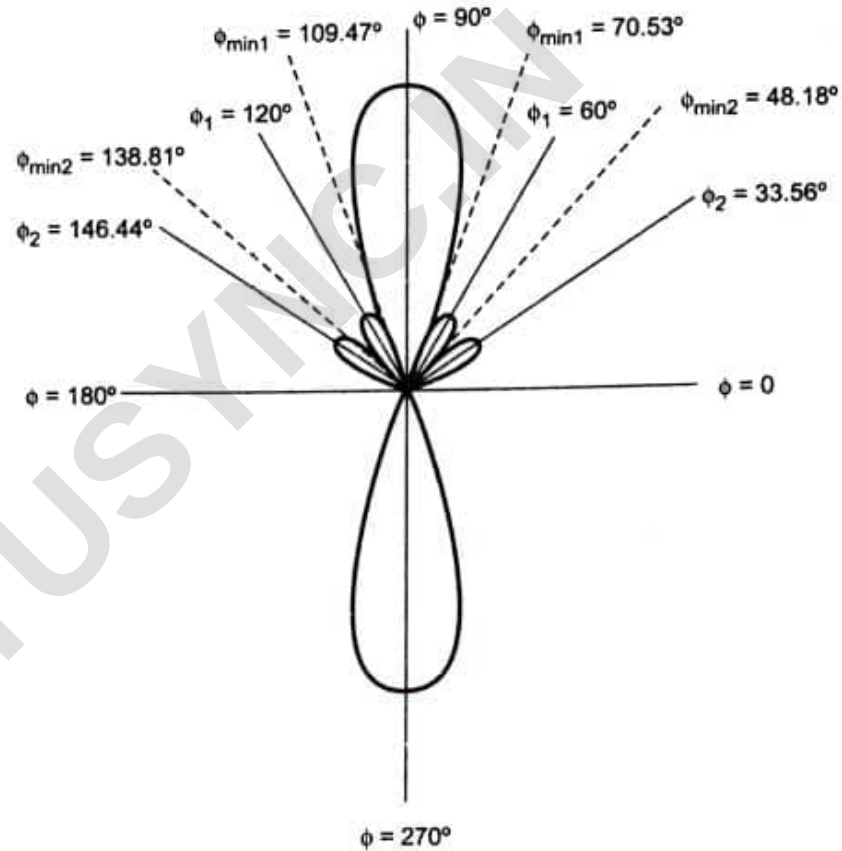
4) Subsidiary maxima (side lobes) : The direction of side lobes is given by,

$$\phi = \cos^{-1} \left[\pm \frac{\lambda (2m+1)}{2nd} \right] \text{ where } m = 1, 2, 3, \dots$$

$$\text{For } m = 1, \phi_1 = \cos^{-1} \left[\pm \frac{\lambda (2+1)}{(2)(6) \left(\frac{\lambda}{2} \right)} \right] = \cos^{-1} \left[\pm \frac{3}{6} \right] = 60^\circ \text{ and } 120^\circ$$

$$\text{For } m = 2, \phi_2 = \cos^{-1} \left[\pm \frac{\lambda (4+1)}{(2)(6) \left(\frac{\lambda}{2} \right)} \right] = \cos^{-1} \left[\pm \frac{5}{6} \right] = 33.56^\circ \text{ and } 146.44^\circ$$

Hence the field pattern is as shown in the Fig. 2.12.4.



ii) The half power bandwidth is given by,

$$\text{HPBW} = 57.3 \left(\frac{1}{\left(\frac{L}{\lambda} \right)} \right)$$

But $L = (n-1)d = (6-1)\frac{\lambda}{2} = \frac{5\lambda}{2}$

$$\therefore \text{HPBW} = 57.3 \left[\frac{1}{\left(\frac{5\lambda / 2}{\lambda} \right)} \right] = 22.92^\circ$$

iii) Hence $\text{FNBW} = 2 (\text{HPBW}) = 2 (22.92^\circ) = 45.84^\circ$

Example 2.12.8 Find the minimum spacing between the elements in a broadside array of 10 isotropic radiators to have directivity of 7 dB.

Solution : Given :

$$G_{Dmax} = 7 \text{ dB}$$

$$n = \text{Number of elements} = 10$$

By definition,

$$G_{Dmax} = 10 \log_{10}[G_{Dmax}]$$

$$\therefore 7 = 10 \log_{10}[G_{Dmax}]$$

$$\therefore G_{Dmax} = 5.0118$$

By formula, the directivity for the broadside array is given by,

$$G_{Dmax} = 2 \left(\frac{L}{\lambda} \right) = 2 \left(\frac{nd}{\lambda} \right)$$

$$\therefore 5.0118 = 2 \left(\frac{10 \times d}{\lambda} \right)$$

$$\therefore d = 0.25 \lambda$$

Hence to achieve directivity of 7 dB with a broadside array of 10 isotropic radiator, the minimum distance between the elements must be 0.25λ .

Example 2.12.5 For an 8 element array with equal spacing of 0.7λ and fed with equal amplitude and phase, find the approximate gain and FNBW.

Solution : As the antennas are fed with currents of equal magnitude and phase, it is nothing but broadside array of linear elements.

For given array,

$$n = \text{Number of elements} = 8$$

$$d = \text{Spacing} = 0.7 \lambda$$

For broadside array, the maximum directional gain is given by,

$$G_{D\max} = 2n \left(\frac{d}{\lambda} \right) = 2(8) \left[\frac{0.7\lambda}{\lambda} \right] = 11.2$$

For broadside array, the FNBW is given by,

$$\text{FNBW} = \frac{2}{\left(\frac{L}{\lambda} \right)} = \frac{2}{\frac{(n-1)d}{\lambda}} = \frac{2\lambda}{(n-1)d}$$

$$\dots \because L = (n-1)d$$

$$\therefore \text{FNBW} = \frac{2\lambda}{(8-1)(0.7\lambda)} = 0.4081 \text{ rad} = 23.383^\circ$$

Example 2.12.7 A broadside array consists four identical half wave dipoles spaced 50 cm apart. If the wavelength is 0.1 m and each element carries r.f. current of equal magnitude of 0.25 A and same phase, calculate power radiated and half power beamwidth of the major lobe.

Solution :

Given :

n = Number of elements = 4

λ = Wavelength = 0.1 m

d = Spacing between any two elements = 50 cm = 0.5 m

I = Current = 0.25 A

Solution :

1) Power radiated is given by,

$$P_{\text{rad}} = n(I^2 R_{\text{rad}})$$

where R_{rad} = Radiation resistance of antenna i.e. half wave diople = 73Ω

$$\therefore P_{\text{rad}} = 4(0.25)^2(73) = 18.25 \text{ W}$$

2) The length of the array in meter is given by,

$$L = nd = 4(0.5) = 2 \text{ m}$$

Hence the bandwidth between first nulls is given by,

$$\text{BWFN} = \frac{2\lambda}{L} = \frac{2 \times 0.1}{2} = 0.1 \text{ rad or } 5.73^\circ$$

Hence half power beamwidth is given by,

$$\text{HPBW} = \frac{\text{BWFN}}{2} = \frac{0.1}{2} = 0.05 \text{ rad or } 2.865^\circ$$

Table: Summary of n element uniform linear end-fire array

Sr. No.	Parameter	Expressions
1.	Direction of Major Lobe	$\phi_{\max} = 0^\circ$
2.	Magnitude of Major Lobe	n
3.	Direction of Minor Lobe	$\phi_{\min} = \cos^{-1} \left[1 - \frac{m\lambda}{nd} \right]$ where m = 1, 2, 3, ...
4.	Direction of Subsidiary Maxima (or Side Lobe)	$\phi = \cos^{-1} \left[1 - \frac{(2m+1)\lambda}{2nd} \right]$ where m = 1, 2, 3, ...
5.	Beamwidth of Major Lobe (BWFN)	$BWFN = \pm 2 \sqrt{\frac{2m}{\left(\frac{L}{\lambda}\right)}} \text{ rad} = \pm 114.6 \sqrt{\frac{2m}{\left(\frac{L}{\lambda}\right)}} \text{ degree}$
6.	Half Power Beam Width (HPBW)	$HPBW = \pm \sqrt{\frac{2m}{\left(\frac{L}{\lambda}\right)}} \text{ rad} = \pm 57.3 \sqrt{\frac{2m}{\left(\frac{L}{\lambda}\right)}} \text{ degree},$ $m = 1, 2, 3, \dots$
7.	Directivity	$D = 4 \left(\frac{L}{\lambda} \right)$

Module 4: Chapter 2

Electric Dipoles:

- Introduction
- Short Electric Dipole
- Fields of a Short Dipole (General and Far Field Analyses)
- Radiation Resistance of a Short Dipole
- Thin Linear Antenna (Field Analyses)
- Radiation Resistances of $\lambda/2$ Antenna

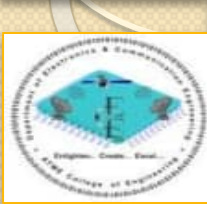
Suggested Reading

Text 3: “Antennas and Wave Propagation”

John D. Krauss, Ronald J Marhefka and Ahmad S Khan, 4th
Special Indian Edition , McGraw- Hill Education Pvt. Ltd., 2010.

Text 3:

Sections: 6.1 -6.6



Introduction

The Short Electric Dipole

Since any linear antenna may be considered as consisting of a large number of very short conductors connected in series, it is of interest to examine first the radiation properties of short conductors. From a knowledge of properties of short conductors, we can then proceed to a study of long linear conductors such as are commonly employed in practice.

A short linear conductor is often called a *short dipole*. In the following discussion, a short dipole is always of finite length even though it may be very short. If the dipole is vanishingly short, it is an infinitesimal dipole.

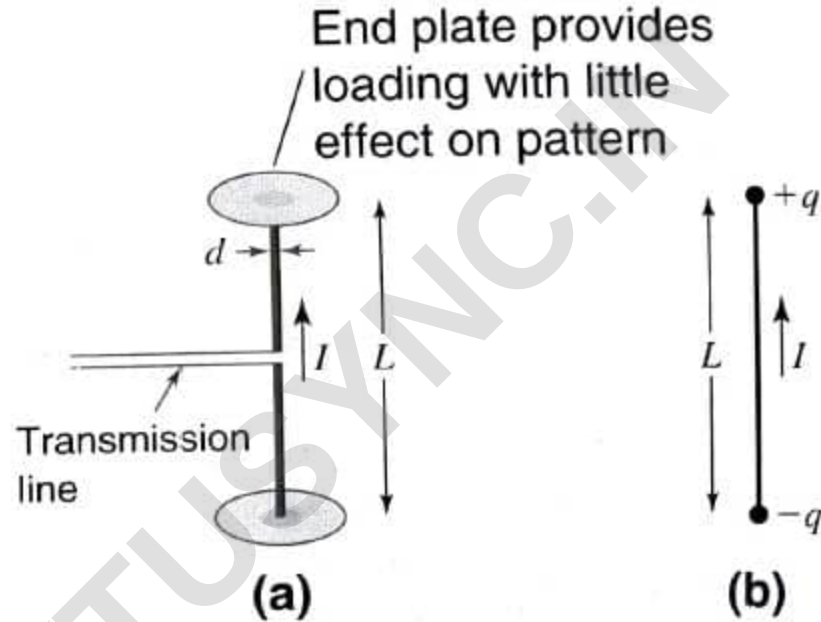


Figure 6-1 A short dipole antenna (a) and its equivalent (b).

Let us consider a short dipole such as shown in Fig. 6-1a. The length L is very short compared to the wavelength ($L \ll \lambda$). Plates at the ends of the dipole provide capacitive loading. The short length and the presence of these plates result in a uniform current I along the entire length L of the dipole. The dipole may be energized by a balanced transmission line, as shown. It is assumed that the transmission line does not radiate and, therefore, its presence will be disregarded. Radiation from the end plates is also considered to be negligible. The diameter d of the dipole is small compared to its length ($d \ll L$). Thus, for purposes of analysis we may consider that the short dipole appears as in Fig. 6-1b. Here it consists simply of a thin conductor of length L with a uniform current I and point charges q at the ends. The current and charge are related by

$$\frac{dq}{dt} = I \quad (1)$$

The Fields of a Short Dipole

Let us now proceed to find the fields everywhere around a short dipole. Let the dipole of length L be placed coincident with the z axis and with its center at the origin as in Fig. 6-2. The relation of the electric field components, E_r , E_θ and E_ϕ , is then as shown. It is assumed that the medium surrounding the dipole is air or vacuum.

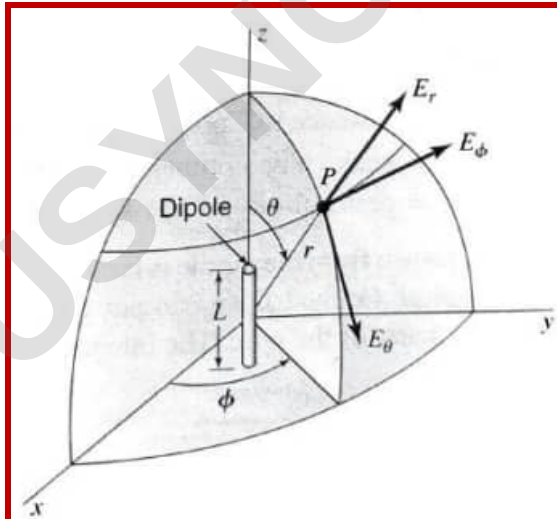
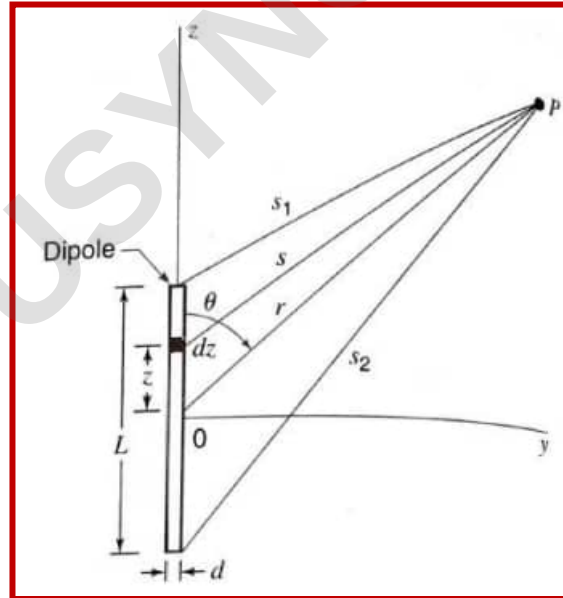


Figure 6-2 Relation of dipole to coordinates.

Retardation Effect

In dealing with antennas or radiating systems, the propagation time is a matter of great importance. Thus, if a current is flowing in the short dipole of Fig. 6–3, the effect of the current is not felt instantaneously at the point P , but only after an interval equal to the time required for the disturbance to propagate over the distance r .



Accordingly, instead of writing the current I as¹

$$I = I_0 e^{j\omega t} \quad (1)$$

which implies instantaneous propagation of the effect of the current, we introduce the propagation (or retardation) time as done by Lorentz and write

$$[I] = I_0 e^{j\omega[t-(r/c)]} \quad (2)$$

where $[I]$ is called the *retarded current*. Specifically, the retardation time r/c results in a *phase retardation* $\omega r/c = 2\pi f r/c$ radians $= 360^\circ f r/c$. The brackets may be added as in (2) to indicate explicitly that the effect of the current is retarded.

Equation (2) is a statement of the fact that the disturbance at a time t and at a distance r from a current element is caused by a current $[I]$ that occurred at an earlier time $t - r/c$. The time difference r/c is the interval required for the disturbance to travel the distance r , where c is the velocity of light ($= 300 \text{ Mm s}^{-1}$).

Electric and magnetic fields can be expressed in terms of vector and scalar potentials. Since we will be interested not only in the fields near the dipole but also at distances which are large compared to the wavelength, we must use *retarded potentials*, i.e., expressions involving $t - r/c$. For a dipole located as in Fig. 6-2 or Fig. 6-3a, the retarded vector potential of the electric current has only one component, namely, A_z . Its value is

$$A_z = \frac{\mu_0}{4\pi} \int_{-L/2}^{L/2} \frac{[I]}{s} dz \quad (2)$$

where $[I]$ is the retarded current given by

$$[I] = I_0 e^{j\omega[t - (s/c)]} \quad (3a)$$

In (3) and (3a),

z = distance to a point on the conductor

I_0 = peak value in time of current (uniform along dipole)

μ_0 = permeability of free space = $4\pi \times 10^{-7} \text{ H m}^{-1}$

If the distance from the dipole is large compared to its length ($r \gg L$) and if the wavelength is large compared to the length ($\lambda \gg L$), we can put $s = r$ and neglect the phase differences of the field contributions from different parts of the wire. The integrand in (3) can then be regarded as a constant, so that (3) becomes

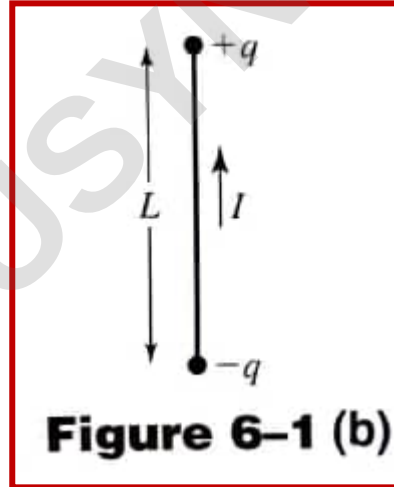
$$A_z = \frac{\mu_0 L I_0 e^{j\omega[t-(r/c)]}}{4\pi r}$$

(4)

Since the region of charge in the case of the dipole being considered is confined to the points at the ends as in Fig. 6-1*b*,

The retarded scalar potential V of a charge distribution is

$$V = \frac{1}{4\pi\epsilon_0} \left\{ \frac{[q]}{s_1} - \frac{[q]}{s_2} \right\} \quad \begin{aligned} \epsilon_0 &= \text{permittivity or dielectric constant of free space} \\ &= 8.85 \times 10^{-12} \text{ F m}^{-1} \end{aligned} \quad (7)$$



$$\frac{dq}{dt} = I \quad (1) \Rightarrow [q] = \int [I] dt$$

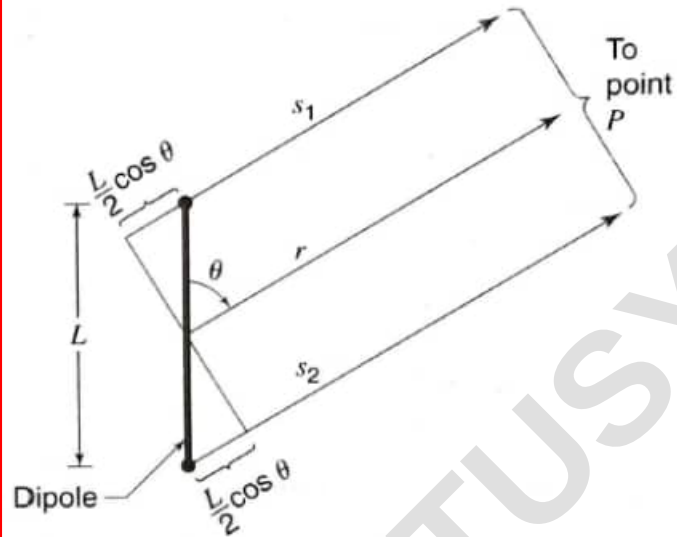
$$[I] = I_0 e^{j\omega[t-(s/c)]} \quad (3a)$$

From (6-1-1) and (3a),

$$[q] = \int [I] dt = I_0 \int e^{j\omega[t-(s/c)]} dt = \frac{[I]}{j\omega} \quad (8)$$

Substituting (8) into (7),

$$V = \frac{I_0}{4\pi\epsilon_0 j\omega} \left[\frac{e^{j\omega[t-(s_1/c)]}}{s_1} - \frac{e^{j\omega[t-(s_2/c)]}}{s_2} \right] \quad (9)$$



Referring to Fig. 6-3b, when $r \gg L$, the lines connecting the ends of the dipole and the point P may be considered as parallel so that

$$s_1 = r - \frac{L}{2} \cos \theta \quad (10)$$

and

$$s_2 = r + \frac{L}{2} \cos \theta \quad (11)$$

Figure 6-3b Relations for short dipole when $r \gg L$.

Substituting (10) and (11) into (9), it may be shown that the fields of a short electric dipole are:

$$\text{Electric fields of short dipole} \quad E_r = \frac{I_0 L \cos \theta e^{j\omega[t-(r/c)]}}{2\pi \epsilon_0} \left(\frac{1}{cr^2} + \frac{1}{j\omega r^3} \right) \quad \text{General case} \quad (12)$$

$$E_\theta = \frac{I_0 L \sin \theta e^{j\omega[t-(r/c)]}}{4\pi \epsilon_0} \left(\frac{j\omega}{c^2 r} + \frac{1}{cr^2} + \frac{1}{j\omega r^3} \right) \quad (13)$$

In obtaining (12) and (13) the relation was used that $\mu_0 \epsilon_0 = 1/c^2$, where c = velocity of light.

$$\text{Magnetic fields of short dipole} \quad |\mathbf{H}| = H_\phi = \frac{I_0 L \sin \theta e^{j\omega[t-(r/c)]}}{4\pi} \left(\frac{j\omega}{cr} + \frac{1}{r^2} \right) \quad \text{General case} \quad (15)$$

$$H_r = H_\theta = 0 \quad (16)$$

Thus, the fields from the dipole have only three components E_r , E_θ and H_ϕ . The components E_ϕ , H_r and H_θ are everywhere zero.

When r is very large, the terms in $1/r^2$ and $1/r^3$ in (12), (13), and (15) can be neglected in favor of the terms in $1/r$. Thus, in the *far field* E_r is negligible, and we have effectively only two field components, E_θ and H_ϕ , given by

$$\text{Electric and magnetic fields of short dipole } E_\theta = \frac{j\omega I_0 L \sin \theta e^{j\omega[t-(r/c)]}}{4\pi \epsilon_0 c^2 r} = j \frac{I_0 \beta L}{4\pi \epsilon_0 c r} \sin \theta e^{j\omega[t-(r/c)]} \quad \text{Far-field case} \quad (17)$$

$$H_\phi = \frac{j\omega I_0 L \sin \theta e^{j\omega[t-(r/c)]}}{4\pi c r} = j \frac{I_0 \beta L}{4\pi r} \sin \theta e^{j\omega[t-(r/c)]} \quad (18)$$

Taking the ratio of E_θ to H_ϕ as given by (17) and (18), we obtain

$$\frac{E_\theta}{H_\phi} = \frac{1}{\epsilon_0 c} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.7 \, \Omega \quad \text{Impedance of space} \quad (19)$$

This is the *intrinsic impedance of free space* (a pure resistance). It is a very important constant.

Comparing (17) and (18) we note that E_θ and H_ϕ are in time phase in the far field. We note also that the field patterns of both are proportional to $\sin \theta$. The pattern is independent of ϕ , so that the space pattern is doughnut-shaped, being a figure-of-revolution of the pattern in Fig. 6-4a about the axis of the dipole.

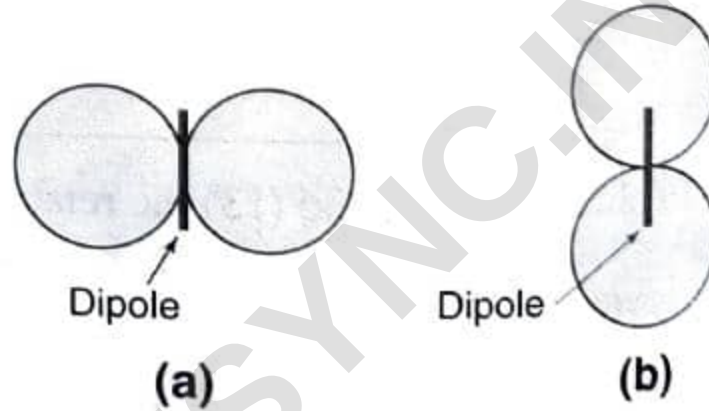


Figure 6-4 Near- and far-field patterns of E_θ and H_ϕ components for short dipole (a) and near-field pattern of E_r component (b).

Let us now consider the situation at very low frequencies. This will be referred to as the *quasi-stationary*, or dc case. Since from

$$[I] = I_0 e^{j\omega[t-(r/c)]} = j\omega[q]$$

(12) and (13) can be rewritten as

$$E_r = \frac{[q]L \cos \theta}{2\pi \epsilon_0} \left(\frac{j\omega}{cr^2} + \frac{1}{r^3} \right) \quad (21)$$

and

$$E_\theta = \frac{[q]L \sin \theta}{4\pi \epsilon_0} \left(-\frac{\omega^2}{c^2 r} + \frac{j\omega}{cr^2} + \frac{1}{r^3} \right) \quad (22)$$

The magnetic field is given by (15) as

$$H_\phi = \frac{[I]L \sin \theta}{4\pi} \left(\frac{j\omega}{cr} + \frac{1}{r^2} \right) \quad (23)$$

At low frequencies, ω approaches zero so that the terms with ω in the numerator can be neglected. As $\omega \rightarrow 0$, we also have

$$[q] = q_0 e^{j\omega[t-(r/c)]} = q_0 \quad (24)$$

and

$$[I] = I_0 \quad (25)$$

Thus, for the quasi-stationary, or dc, case, the field components become from (21), (22) and (23)

$$E_r = \frac{q_0 L \cos \theta}{2\pi \epsilon_0 r^3} \quad (26)$$

$$\text{Electric and magnetic fields of short dipole} \quad E_\theta = \frac{q_0 L \sin \theta}{4\pi \epsilon_0 r^3} \quad \text{Low-frequency case} \quad (27)$$

$$H_\phi = \frac{I_0 L \sin \theta}{4\pi r^2} \quad (28)$$

Table 6-1 Fields of a short electric dipole[†]

Component	General expression	Far field	Quasi-stationary
E_r	$\frac{[I]L \cos \theta}{2\pi \epsilon_0} \left(\frac{1}{cr^2} + \frac{1}{j\omega r^3} \right)$	0	$\frac{q_0 L \cos \theta}{2\pi \epsilon_0 r^3}$
E_θ	$\frac{[I]L \sin \theta}{4\pi \epsilon_0} \left(\frac{j\omega}{c^2 r} + \frac{1}{cr^2} + \frac{1}{j\omega r^3} \right)$	$\frac{[I]L j\omega \sin \theta}{4\pi \epsilon_0 c^2 r} = \frac{j60\pi [I] \sin \theta}{r} \frac{L}{\lambda}$	$\frac{q_0 L \sin \theta}{4\pi \epsilon_0 r^3}$
H_ϕ	$\frac{[I]L \sin \theta}{4\pi} \left(\frac{j\omega}{cr} + \frac{1}{r^2} \right)$	$\frac{[I]L j\omega \sin \theta}{4\pi cr} = \frac{j[I] \sin \theta}{2r} \frac{L}{\lambda}$	$\frac{I_0 L \sin \theta}{4\pi r^2}$

CBCS SCHEME

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15EC71

Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019

Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing
ONE full question from each module.

Module-4

- 7 a. Derive an expression and draw the field pattern for an array of 2 isotropic point sources with same amplitude and phase spaced $\lambda/2$ apart. (06 Marks)
- b. Show that the radiation resistance of $\lambda/2$ antenna is 73Ω . (06 Marks)
- c. A source has a radiation intensity power pattern given by $U = U_m \sin^2 \theta$ for $0 \leq \theta \leq \pi$; $0 \leq \phi \leq 2\pi$. Find the total power and directivity. Draw pattern. (04 Marks)

OR

- 8 a. Derive the expressions for the far field components of short dipole. (06 Marks)
- b. Explain the principle of pattern multiplication with an example. (06 Marks)
- c. A source has a cosine radiation intensity pattern given by $U = U_m \cos \theta$ for $0 \leq \theta \leq \pi/2$ and $0 \leq \phi \leq 2\pi$. Find the total power and directivity. (04 Marks)

Radiation Resistance of Short Electric Dipole

CBCS SCHEME

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15EC71

Seventh Semester B.E. Degree Examination, Dec.2019/Jan.2020 Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-4

- 7 a. Prove that directivity for a source with unidirectional pattern of $U_m \cos^n \theta$, where 'n' can be any number, can be expressed as $D = 2(n + 1)$. (06 Marks)
- b. Obtain field expression of two isotropic point sources of same amplitude and phase. (10 Marks)
- OR
- 8 a. State and explain power theorem. (06 Marks)
- b. Derive an expression for radiation resistance of short electric dipole. (10 Marks)

CBCS SCHEME

USN

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15EC71

Seventh Semester B.E. Degree Examination, June/July 2019 Microwave and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-4

- 7 a. Obtain the fields pattern for two point source situated symmetrically with respect to the origin. Two sources are feed with equal amplitude and equal phase signals. Assume distance between two sources = $\lambda/2$. (08 Marks)
- b. Derive the expression for radiation resistance of short electric dipole. (08 Marks)

OR

- 8 a. Derive an array factor expression in case of linear array of 'n' isotropic point source of equal amplitude and spacing. (08 Marks)
- b. Obtain the expression for field of dipole in general for the case of thin linear antenna. (08 Marks)

Radiation Resistance of Short Electric Dipole

Let us now calculate the radiation resistance of the short dipole of Fig. 6-1b. This may be done as follows. The Poynting vector of the far field is integrated over a large sphere to obtain the total power radiated. This power is then equated to $I^2 R$ where I is the rms current on the dipole and R is a resistance, called the radiation resistance of the dipole.

The average Poynting vector is given by

$$S = \frac{1}{2} \text{Re}(E \times H^*) \quad (1)$$

The far-field components are E_θ and H_ϕ so that the radial component of the Poynting vector is

$$S_r = \frac{1}{2} \text{Re} E_\theta H_\phi^* \quad (2)$$

where E_θ and H_ϕ^* are complex.

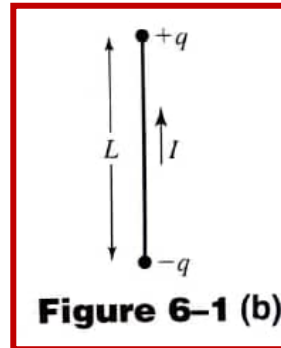


Figure 6-1 (b)

The far-field components are related by the intrinsic impedance of the medium. Hence,

$$E_{\theta} = H_{\phi} Z = H_{\phi} \sqrt{\frac{\mu}{\epsilon}} \quad (3)$$

Thus, (2) becomes

$$S_r = \frac{1}{2} \operatorname{Re} Z H_{\phi} H_{\phi}^* = \frac{1}{2} |H_{\phi}|^2 \operatorname{Re} Z = \frac{1}{2} |H_{\phi}|^2 \sqrt{\frac{\mu}{\epsilon}} \quad (4)$$

The total power P radiated is then

$$P = \iint S_r ds = \frac{1}{2} \sqrt{\frac{\mu}{\epsilon}} \int_0^{2\pi} \int_0^{\pi} |H_{\phi}|^2 r^2 \sin \theta d\theta d\phi \quad (5)$$

where the angles are as shown in Fig. 6-2 and $|H_{\phi}|$ is the absolute value of the magnetic field, which from (6-3-18) is

$$|H_{\phi}| = \frac{\omega I_0 L \sin \theta}{4\pi cr} \quad (6)$$

Substituting this into (5), we have

$$P = \frac{1}{32} \sqrt{\frac{\mu}{\epsilon}} \frac{\beta^2 I_0^2 L^2}{\pi^2} \int_0^{2\pi} \int_0^\pi \sin^3 \theta \, d\theta \, d\phi \quad (7)$$

The double integral equals $8\pi/3$ and (7) becomes

$$P = \sqrt{\frac{\mu}{\epsilon}} \frac{\beta^2 I_0^2 L^2}{12\pi} \quad (8)$$

This is the *average* power or rate at which energy is streaming out of a sphere surrounding the dipole. Hence, it is equal to the power radiated. Assuming no losses, it is also equal to the power delivered to the dipole.

Therefore, P must be equal to the square of the rms current I flowing on the dipole times a resistance R_r called the *radiation resistance* of the dipole. Thus,

$$\sqrt{\frac{\mu}{\varepsilon}} \frac{\beta^2 I_0^2 L^2}{12\pi} = \left(\frac{I_0}{\sqrt{2}}\right)^2 R_r \quad (9)$$

Solving for R_r ,

$$R_r = \sqrt{\frac{\mu}{\varepsilon}} \frac{\beta^2 L^2}{6\pi} \quad (10)$$

For air or vacuum $\sqrt{\mu/\varepsilon} = \sqrt{\mu_0/\varepsilon_0} = 377 = 120\pi \Omega$ so that (10) becomes¹

Dipole with uniform current	$R_r = 80\pi^2 \left(\frac{L}{\lambda}\right)^2 = 80\pi^2 L_\lambda^2 = 790 L_\lambda^2 \quad (\Omega)$	Radiation resistance	(11)
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$$P = \sqrt{\frac{\mu}{\epsilon}} \frac{\beta^2 I_0^2 L^2}{12\pi} \quad (8)$$

It has been assumed that with end loading (see Fig. 6-1a) the dipole current is uniform. However, with no end loading the current must be zero at the ends and, if the dipole is short, the current tapers almost linearly from a maximum at the center to zero at the ends, as in Fig. 2-12, with an average value of $\frac{1}{2}$ of the maximum. Modifying (8) for the general case where the current is not uniform on the dipole, the *radiated power* is

$$P = \sqrt{\frac{\mu}{\epsilon}} \frac{\beta^2 I_{av}^2 L^2}{12\pi} \quad (W) \quad (12)$$

where I_{av} = amplitude of *average current* on dipole (peak value in time)

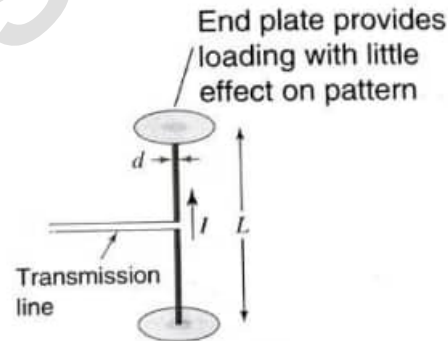


Figure 6-1 (a)

$$P = \sqrt{\frac{\mu}{\varepsilon}} \frac{\beta^2 I_{av}^2 L^2}{12\pi} \quad (\text{W}) \quad (12)$$

The power delivered to the dipole is, as before,

$$P = \frac{1}{2} I_0^2 R_r \quad (\text{W}) \quad (13)$$

where I_0 = amplitude of terminal current of center-fed dipole (peak value in time). Equating the power radiated (12) to the power delivered (13) yields, for free space ($\mu = \mu_0$ and $\varepsilon = \varepsilon_0$), a radiation resistance

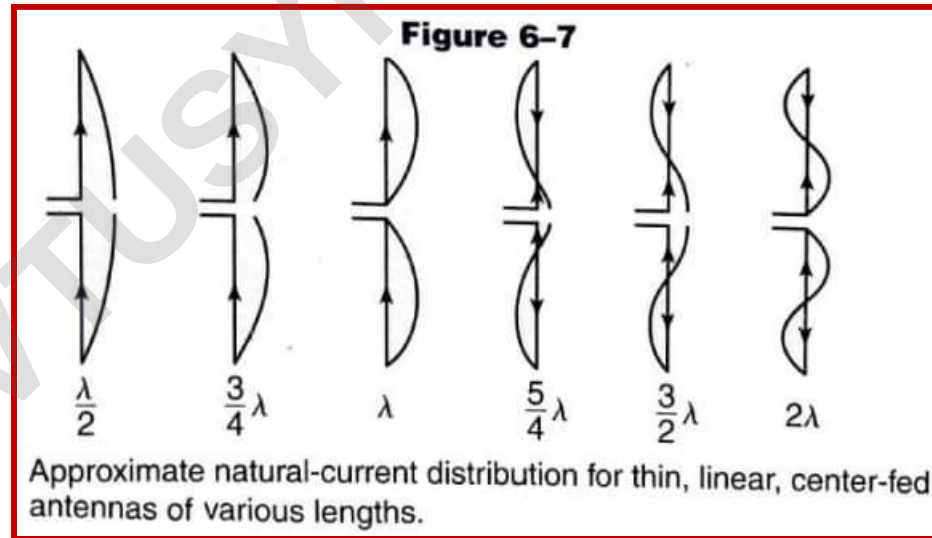
$$R_r = 790 \left(\frac{I_{av}}{I_0} \right)^2 L_\lambda^2 \quad (\Omega) \quad (14)$$

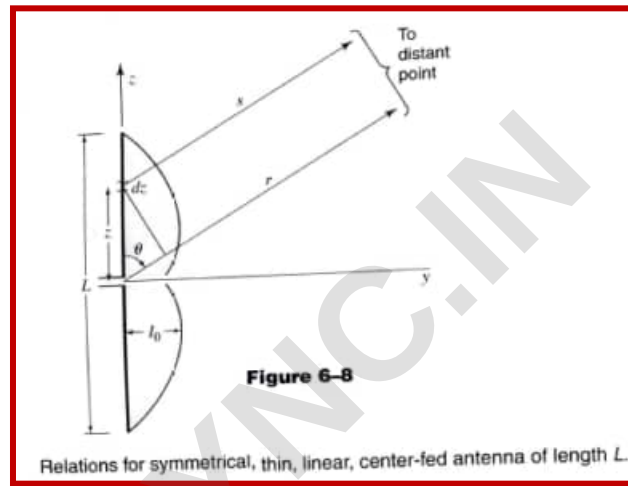
For a short dipole without end loading, we have $I_{av} = \frac{1}{2} I_0$, as noted above, and (14) becomes

$$R_r = 197 L_\lambda^2 \quad (\Omega) \quad (15)$$

The Thin Linear Antenna

In this section expressions for the far-field patterns of thin linear antennas will be developed. It is assumed that the antennas are symmetrically fed at the center by a balanced two-wire transmission line. The antennas may be of any length, but it is assumed that the current distribution is sinusoidal. Current-distribution measurements indicate that this is a good assumption provided that the antenna is thin, i.e., when the conductor diameter is less than, say, $\lambda/100$. Thus, the sinusoidal current distribution approximates the natural distribution on thin antennas. Examples of the approximate natural-current distributions on a number of thin, linear center-fed antennas of different length are illustrated in Fig. 6-7. The currents are in phase over each $\lambda/2$ section and in opposite phase over the next.





Referring to Fig. 6-8, let us now proceed to develop the far-field equations for a symmetrical, thin, linear, center-fed antenna of length L . The retarded value of the current at any point z on the antenna referred to a point at a distance s is

$$[I] = I_0 \sin \left[\frac{2\pi}{\lambda} \left(\frac{L}{2} \pm z \right) \right] e^{j\omega[t - (r/c)]} \quad (1)$$

In (1) the function

$$\sin \left[\frac{2\pi}{\lambda} \left(\frac{L}{2} \pm z \right) \right]$$

is the form factor for the current on the antenna. The expression $(L/2) + z$ is used when $z < 0$ and $(L/2) - z$ is used when $z > 0$.

By regarding the antenna as made up of a series of infinitesimal dipoles of length dz , the field of the entire antenna may then be obtained by integrating the fields from all of the dipoles making up the antenna with the result¹

*Far fields of
center-fed dipole*

$$H_{\phi} = \frac{j[I_0]}{2\pi r} \left[\frac{\cos[(\beta L \cos \theta)/2] - \cos(\beta L/2)}{\sin \theta} \right] \quad (2)$$

$$E_{\theta} = \frac{j60[I_0]}{r} \left[\frac{\cos[(\beta L \cos \theta)/2] - \cos(\beta L/2)}{\sin \theta} \right] \quad (3)$$

where $[I_0] = I_0 e^{j\omega[t-(r/c)]}$ and

$$E_{\theta} = 120\pi H_{\phi} \quad (3a)$$

Equations (2), (3) and (3a) give the far fields H_{ϕ} and E_{θ} of a *symmetrical, center-fed, thin linear antenna of length L* . The shape of the far-field pattern is given by the factor in the brackets. The factors preceding the brackets in (2) and (3) give the instantaneous magnitude of the fields as functions of the antenna current and the distance r .

Radiation Resistance of Lambda/2 Antenna

To find the radiation resistance, the Poynting vector is integrated over a large sphere yielding the power radiated, and this power is then equated to $(I_0/\sqrt{2})^2 R_0$, where R_0 is the radiation resistance at a current maximum point and I_0 is the peak value in time of the current at this point.

The total power P radiated is then

$$P = \iint S_r ds = \frac{1}{2} \sqrt{\frac{\mu}{\varepsilon}} \int_0^{2\pi} \int_0^\pi |H_\phi|^2 r^2 \sin \theta d\theta d\phi$$

$$H_\phi = \frac{j[I_0]}{2\pi r} \left[\frac{\cos[(\beta L \cos \theta)/2] - \cos(\beta L/2)}{\sin \theta} \right]$$

$$P = \frac{15I_0^2}{\pi} \int_0^{2\pi} \int_0^\pi \frac{\{\cos[(\beta L/2) \cos \theta] - \cos(\beta L/2)\}^2}{\sin \theta} d\theta d\phi \quad (1)$$

$$= 30I_0^2 \int_0^\pi \frac{\{\cos[(\beta L/2) \cos \theta] - \cos(\beta L/2)\}^2}{\sin \theta} d\theta \quad (2)$$

Equating the radiated power as given by (2) to $I_0^2 R_0/2$ we have

$$P = \frac{I_0^2 R_0}{2} \quad (3)$$

and

$$R_0 = 60 \int_0^\pi \frac{[\cos[(\beta L/2) \cos \theta] - \cos(\beta L/2)]^2}{\sin \theta} d\theta \quad (4)$$

where the radiation resistance R_0 is referred to the current maximum. In the case of a $\lambda/2$ antenna this is at the center of the antenna or at the terminals of the transmission line (see Fig. 6-7).

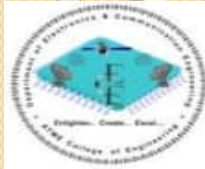
Proceeding with the evaluation of (4) with the aid of the sine integral, $\text{Si}(x)$, and the cosine integral, $\text{Cin}(x)$, it may be shown that the radiation resistance of the $\lambda/2$ antenna is

$$R_r = 30 \text{ Cin}(2\pi) = 30 \times 2.44 = 73 \Omega \quad (5)$$

This is the well-known value for the radiation resistance of a thin, linear, center-fed, $\lambda/2$ antenna with sinusoidal current distribution. The terminal impedance also includes some inductive reactance.

That is,

$$Z = 73 + j42.5 \Omega \quad (6)$$



Thank you